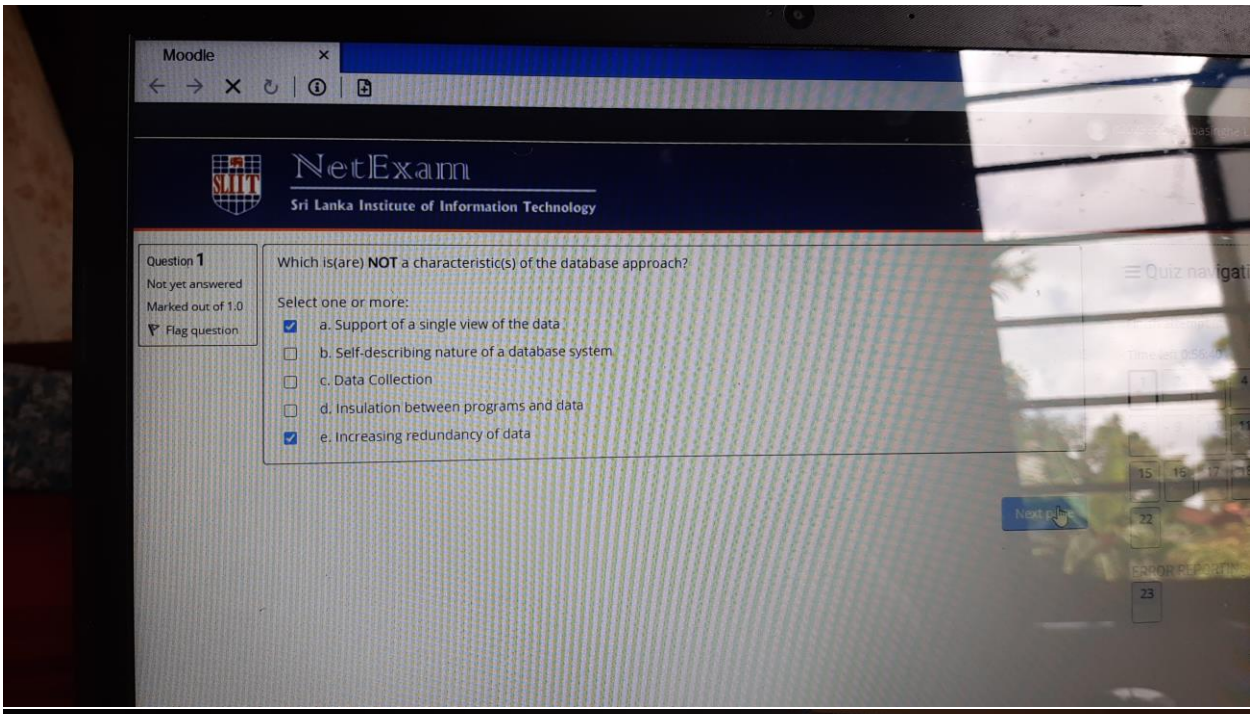
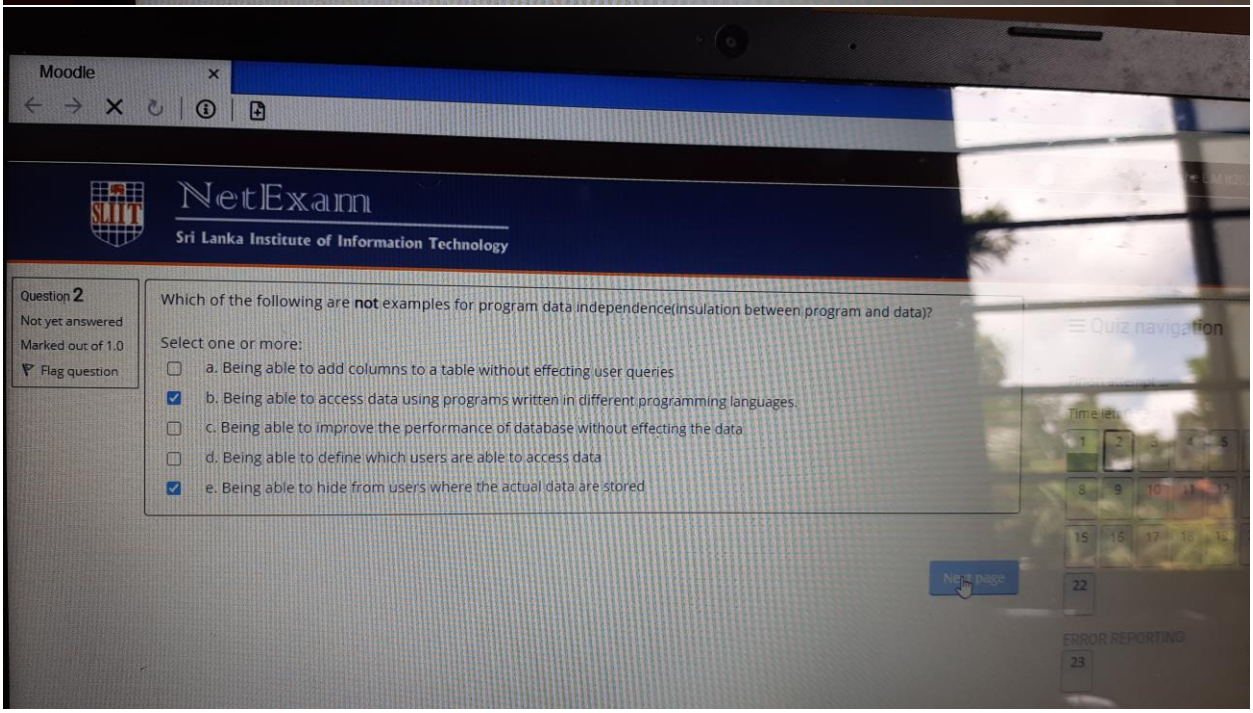
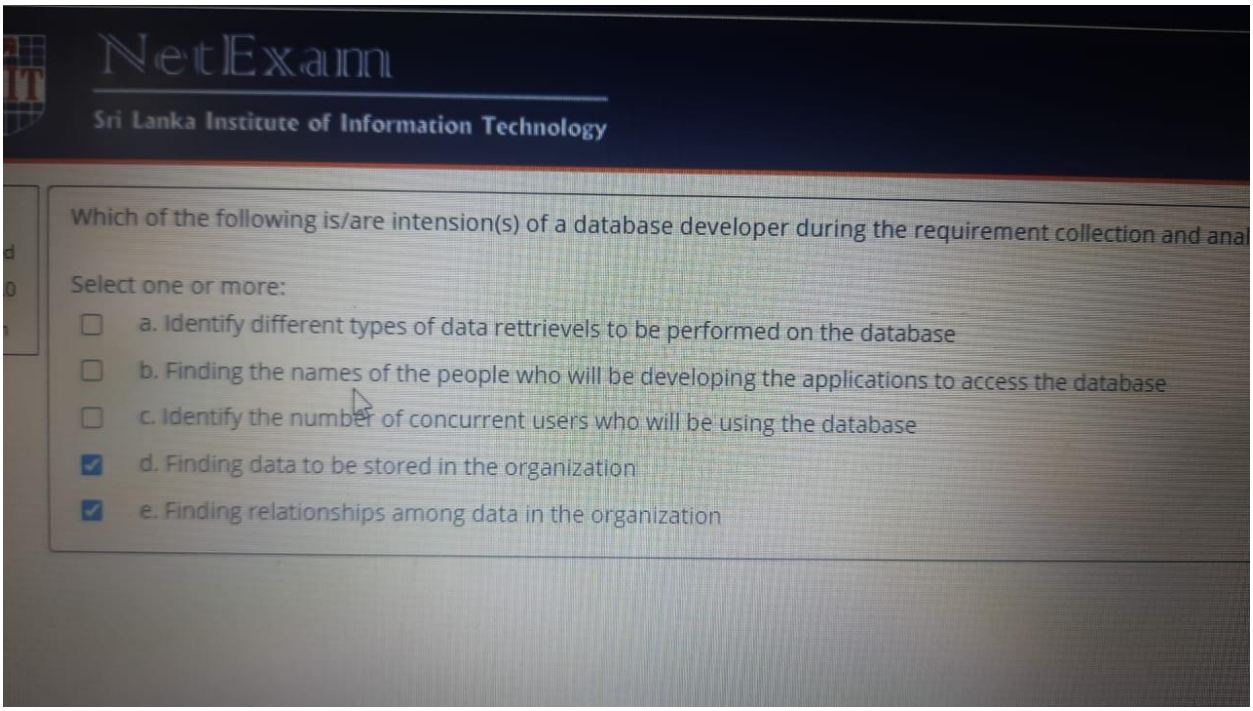
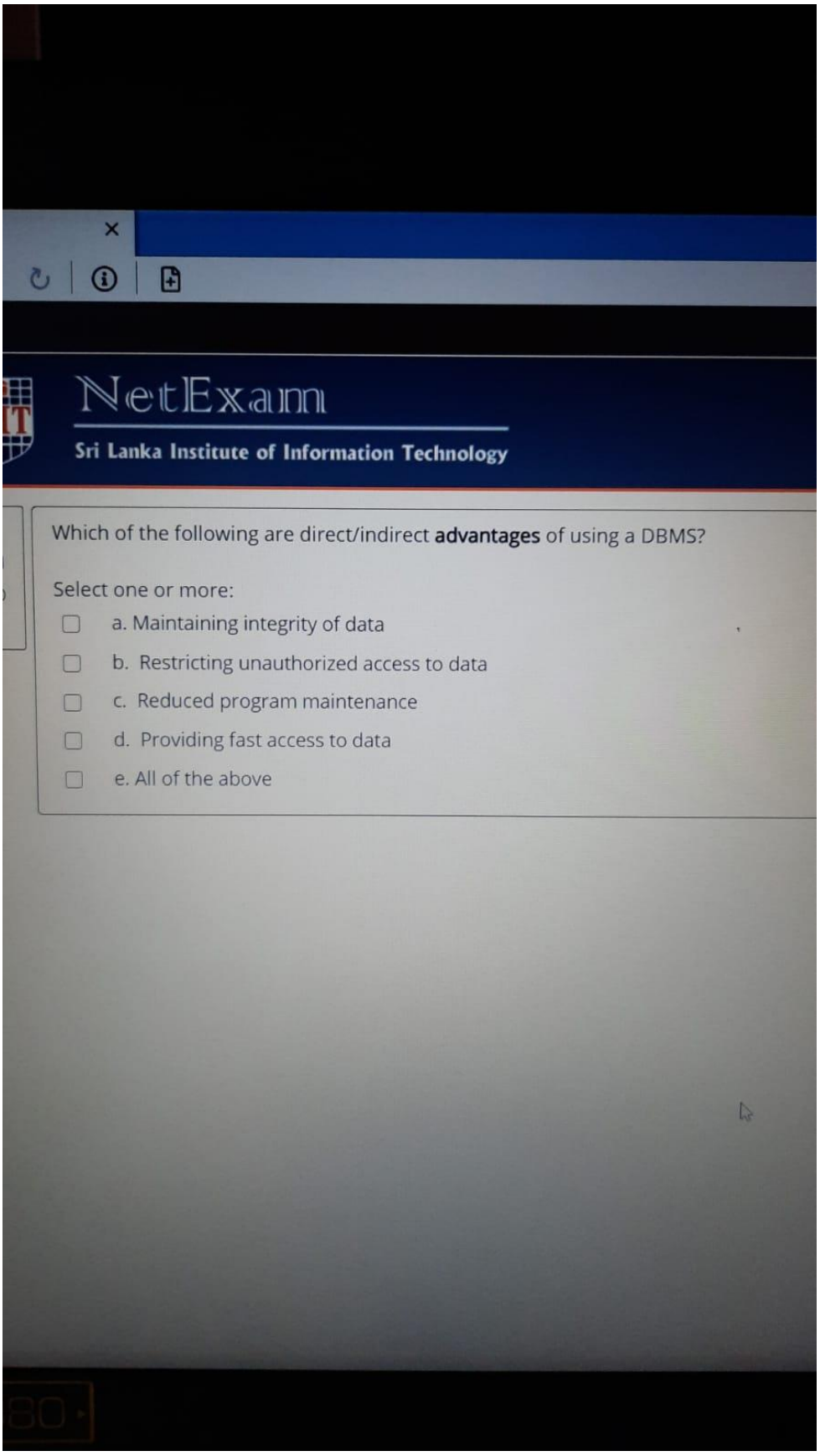

Moodle
NetExam
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Question 1
Not yet answered
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Flag question
Which is(are) NOT a characteristic(s) of the database approach?
Select one or more:
a. Support of a single view of the data
b. Self-describing nature of a database system
c. Data Collection
d. Insulation between programs and data
e. Increasing redundancy of data


Moodle
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Sri Lanka Institute of Information Technology
Question 2
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Flag question
Which of the following are not examples for program data independence(insulation between program and data)?
Select one or more:
a. Being able to add columns to a table without effecting user queries
b. Being able to access data using programs written in different programming languages
c. Being able to improve the performance of database without effecting the data
d. Being able to define which users are able to access data
e. Being able to hide from users where the actual data are stored


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Which of the following is/are intension(s) of a database developer during the requirement collection and anal
Select one or more:
a. Identify different types of data rettrievels to be performed on the database
b. Finding the names of the people who will be developing the applications to access the database
c. Identify the number of concurrent users who will be using the database
d. Finding data to be stored in the organization
e. Finding relationships among data in the organization


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Which of the following are direct/indirect advantages of using a DBMS?
Select one or more:
a. Maintaining integrity of data
b. Restricting unauthorized access to data
c. Reduced program maintenance
d. Providing fast access to data
e. All of the above

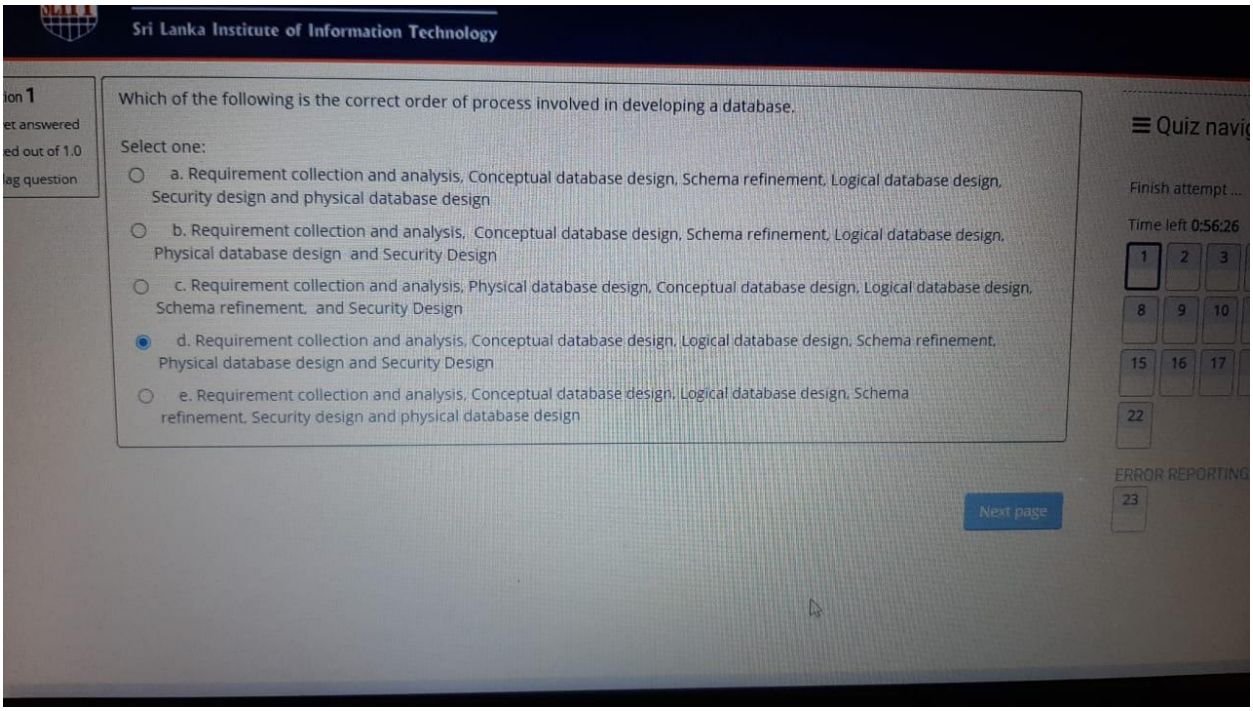
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Which of the following is/are **TRUE** about databases and database management systems?

Select one or more:

- [ ] a. Databases are used to only store complex information
- [ ] b. All databases are computerized
- [ ] c. Defining a database involves specifying the data types, structures and constraints on data
- [ ] d. DBMS is a special purpose software that is capable of structuring, storing and programming data
- [ ] e. Database definitions are stored in a database catalog


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ag question
Which of the following is the correct order of process involved in developing a database.
Select one:
a. Requirement collection and analysis, Conceptual database design, Schema refinement, Logical database design, Security design and physical database design
b. Requirement collection and analysis, Conceptual database design, Schema refinement, Logical database design, Physical database design and Security Design
c. Requirement collection and analysis, Physical database design, Conceptual database design, Logical database design, Schema refinement, and Security Design
d. Requirement collection and analysis, Conceptual database design, Logical database design, Schema refinement, Physical database design and Security Design
e. Requirement collection and analysis, Conceptual database design, Logical database design, Schema refinement, Security design and physical database design
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Which of the following is true related to ISA relationships?

Select one or more:

- [ ] a. The hierarchy in which each entity participates in only one subclass relationship is classified as disjoint
- [ ] b. A sub type of a ISA hierarchy can has its own key
- [ ] c. A shape object with circle, rectangle and triangle as sub types is an example for a total participation
- [ ] d. If subtypes are overlapping the participation constraint must be partial

NetExam

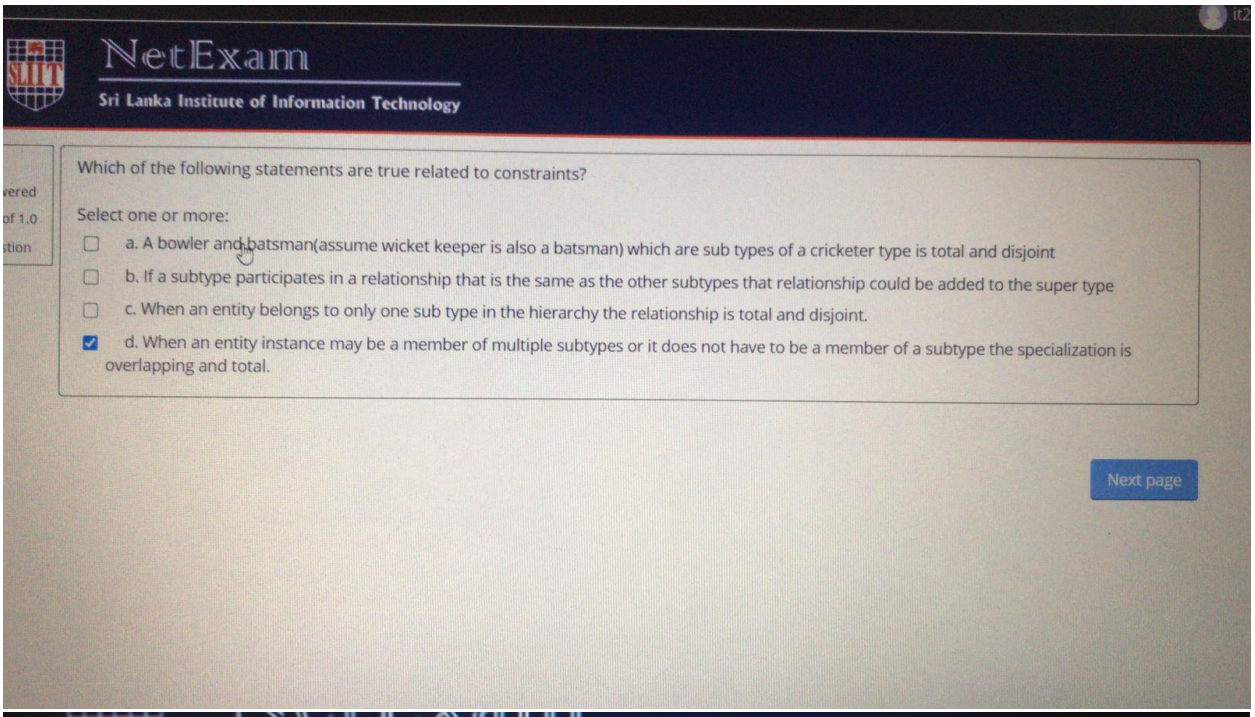
Sri Lanka Institute of Information Technology

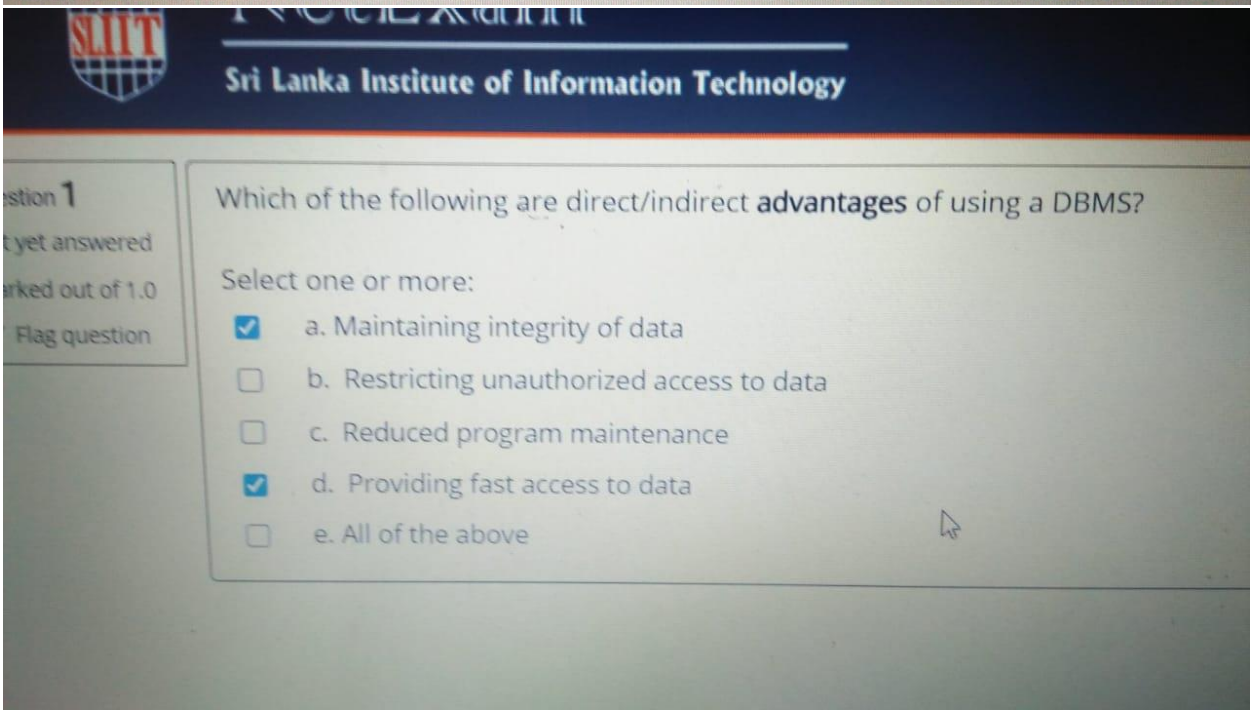
Consider the following activities performed by a database developer while developing a database for a small pharmacy.

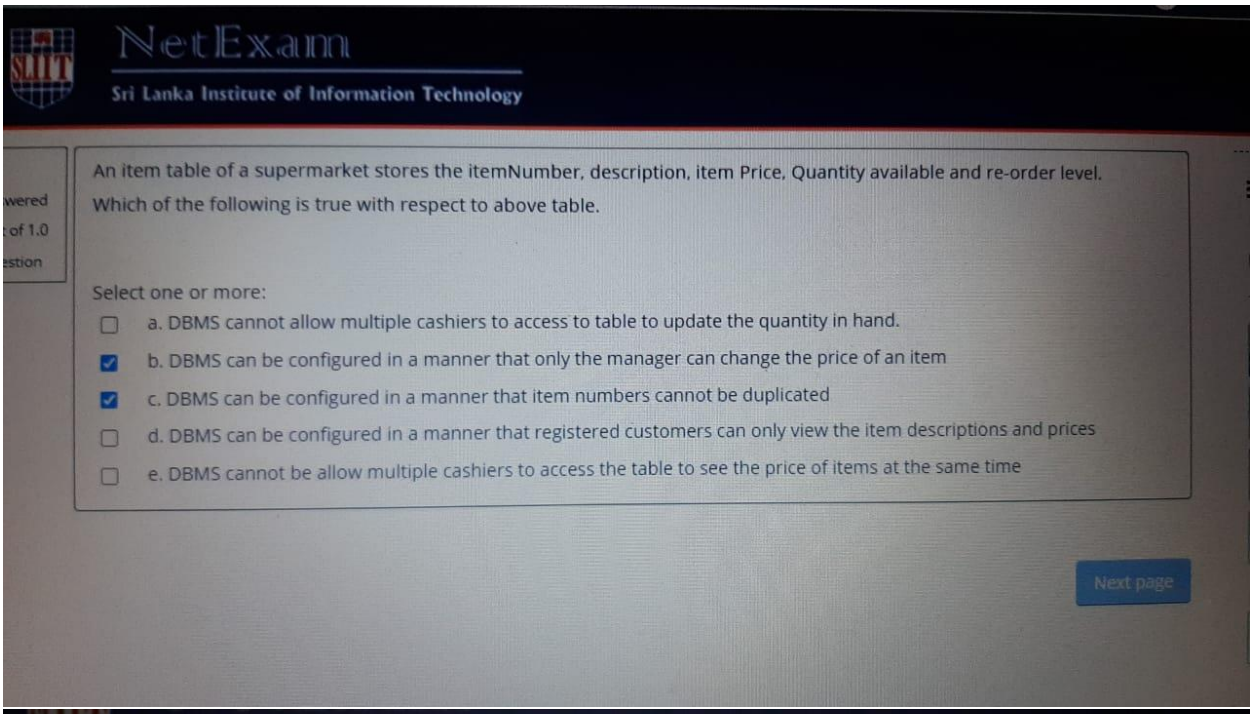
1. Go through the books maintained for recording supplies
2. Identify attributes that determine certain groups of attributes
3. Select a database development software
4. Give access to clerks to enter data

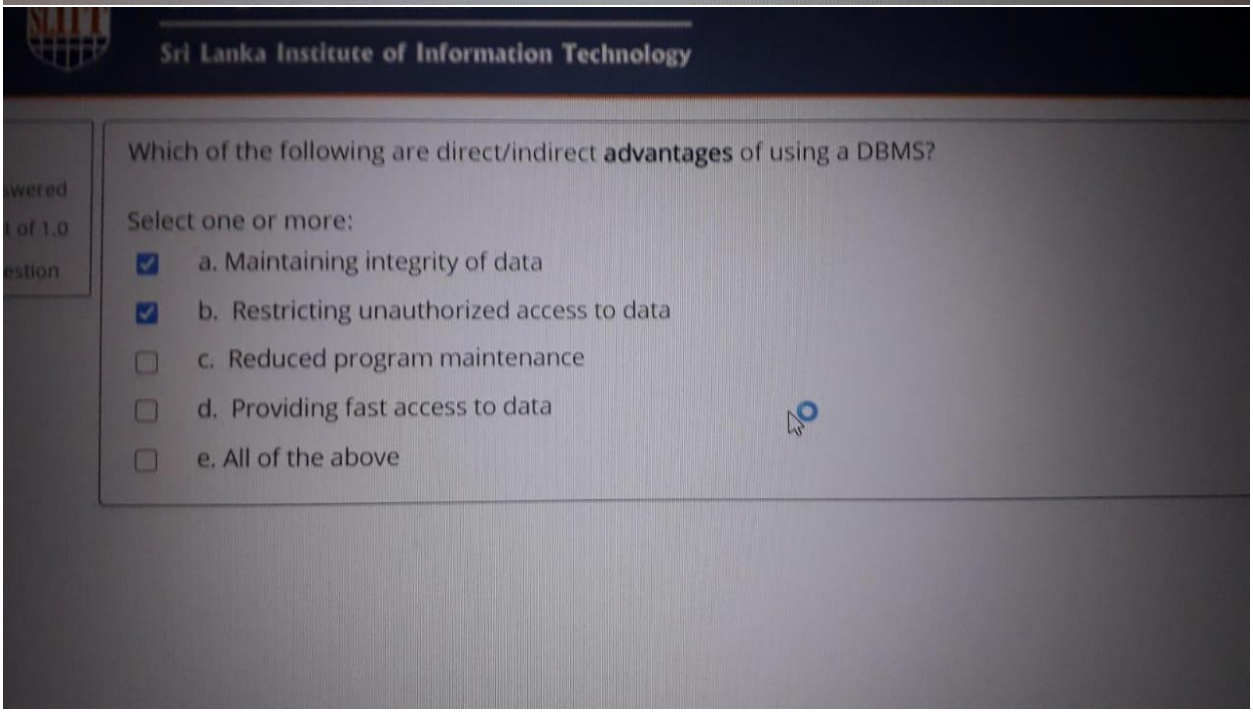
Select one:

- a. 3, 2, 4, 1
- b. 4,1, 3, 2
- c. 2, 3, 4, 1
- d. 1, 3, 2, 4
- e. 1, 2, 3, 4


NetExam
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Which of the following statements are true related to constraints?
Select one or more:
a. A bowler and batsman(assume wicket keeper is also a batsman) which are sub types of a cricketer type is total and disjoint
b. If a subtype participates in a relationship that is the same as the other subtypes that relationship could be added to the super type
c. When an entity belongs to only one sub type in the hierarchy the relationship is total and disjoint.
d. When an entity instance may be a member of multiple subtypes or it does not have to be a member of a subtype the specialization is overlapping and total.
Next page


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Which of the following are direct/indirect advantages of using a DBMS?
Select one or more:
a. Maintaining integrity of data
b. Restricting unauthorized access to data
c. Reduced program maintenance
d. Providing fast access to data
e. All of the above
Flag question


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An item table of a supermarket stores the itemNumber, description, item Price, Quantity available and re-order level.
Which of the following is true with respect to above table.
Select one or more:
a. DBMS cannot allow multiple cashiers to access to table to update the quantity in hand.
b. DBMS can be configured in a manner that only the manager can change the price of an item
c. DBMS can be configured in a manner that item numbers cannot be duplicated
d. DBMS can be configured in a manner that registered customers can only view the item descriptions and prices
e. DBMS cannot be allow multiple cashiers to access the table to see the price of items at the same time
Next page


Sri Lanka Institute of Information Technology
wered
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estion
Which of the following are direct/indirect advantages of using a DBMS?
Select one or more:
a. Maintaining integrity of data
b. Restricting unauthorized access to data
c. Reduced program maintenance
d. Providing fast access to data
e. All of the above

NetExam

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Which of the following situations would you use a database to store data?

Select one or more:

- [x] a. To store information about rooms and customers of a hotel management system
- [x] b. To store student name list of your class which is used by multiple lecturers
- [x] c. To store information of vehicles owned by a vehicle renting company
- [ ] d. To store your 'to do' list
- [ ] e. To store addresses of relatives and friends

Which of the following are **not** examples for program data independence(insulation between program and data)?

Select one or more:

- [ ] a. Being able to access data using programs written in different programming languages.
- [ ] b. Being able to define which users are able to access data
- [ ] c. Being able to improve the performance of database without effecting the data
- [ ] d. Being able to hide from users where the actual data are stored
- [ ] e. Being able to add columns to a table without effecting user queries

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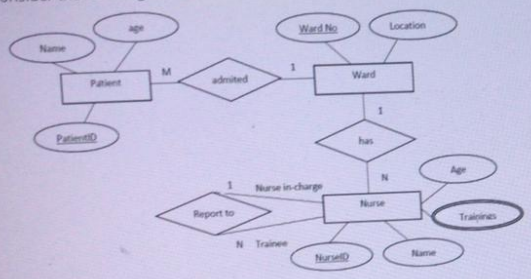
Which of the following statements are true related to constraints?

Select one or more:

- [ ] a. If a subtype participates in a relationship that is the same as the other subtypes that relationship could be added to the super type
- [x] b. When an entity belongs to only one sub type in the hierarchy the relationship is total and disjoint.
- [x] c. When an entity instance may be a member of multiple subtypes or it does not have to be a member of a subtype the specialization is overlapping and total.
- [ ] d. A bowler and batsman(assume wicket keeper is also a batsman) which are sub types of a cricketer type is total and disjoint

Next page

Consider the EER diagram below :

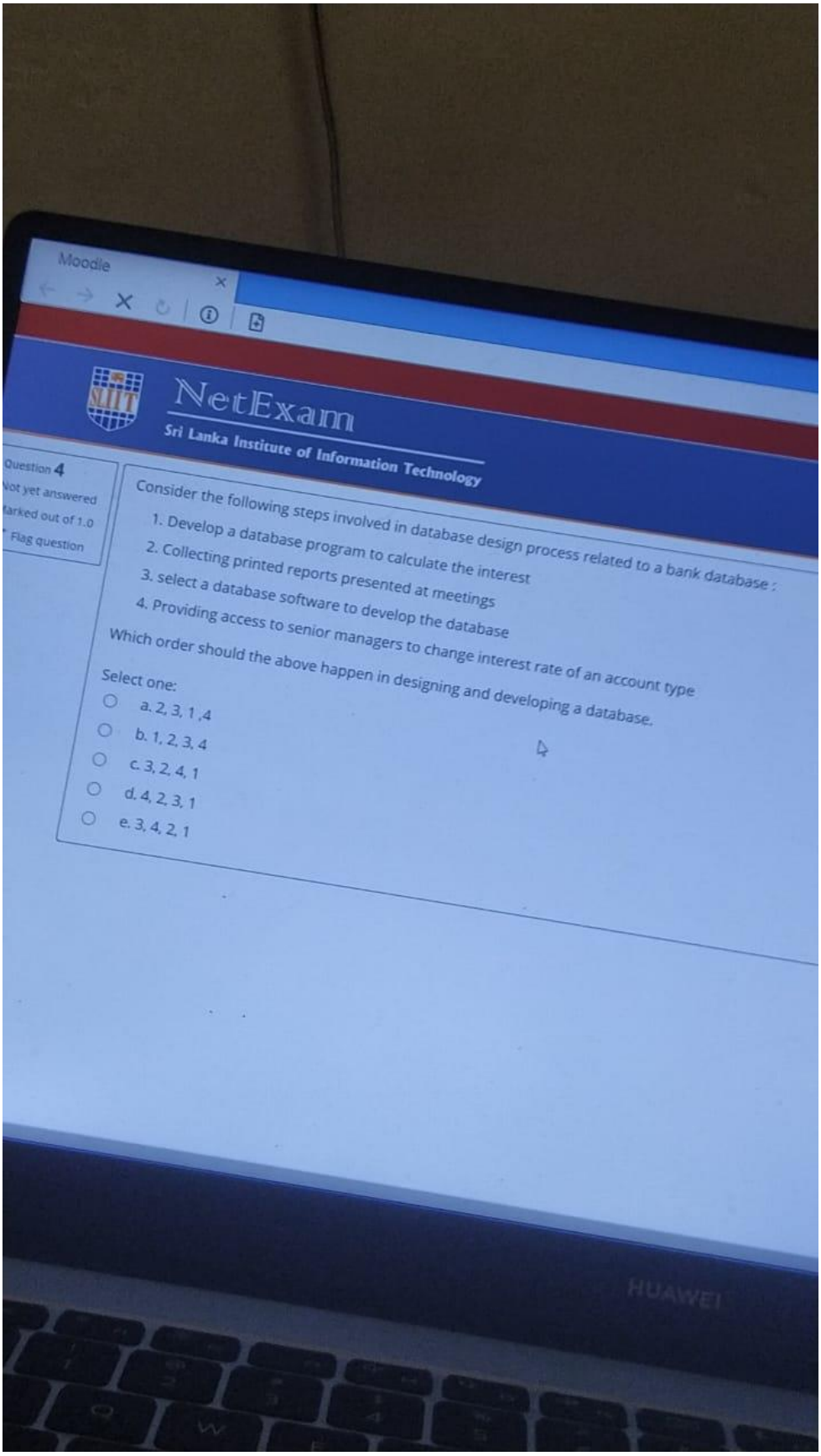


Which of the following statements are true related to the diagram above?

Select one or more:

- [ ] a. No two wards could be at the same location
- [ ] b. There can be wards with no patients
- [ ] c. A trainee nurse may report to multiple nurses
- [ ] d. A nurse only works in only one ward
- [ ] e. All nurses should have at least one training

Next page

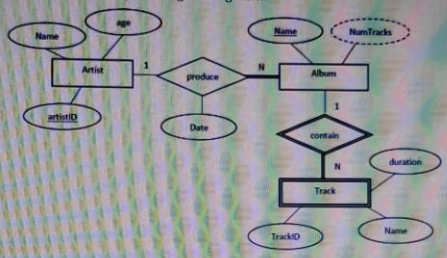

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Question 4
Not yet answered
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Flag question
Consider the following steps involved in database design process related to a bank database :
1. Develop a database program to calculate the interest
2. Collecting printed reports presented at meetings
3. select a database software to develop the database
4. Providing access to senior managers to change interest rate of an account type
Which order should the above happen in designing and developing a database.
Select one:
a. 2, 3, 1 ,4
b. 1, 2, 3, 4
c. 3, 2, 4, 1
d. 4, 2, 3, 1
e. 3, 4, 2, 1
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Question 6
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Flag question

Consider the following EER diagram segment.



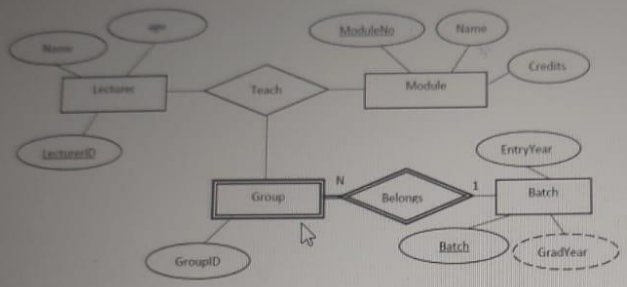
Which of the following is true related to the above diagram?

Select one or more:

- a. There can be multiple tracks with ID 1 in the system
- b. There could be Artists who had not produced any album yet.
- c. An album is produced with multiple artists
- d. Artists name cannot have duplicate values
- e. Number of tracks in an album could be calculated

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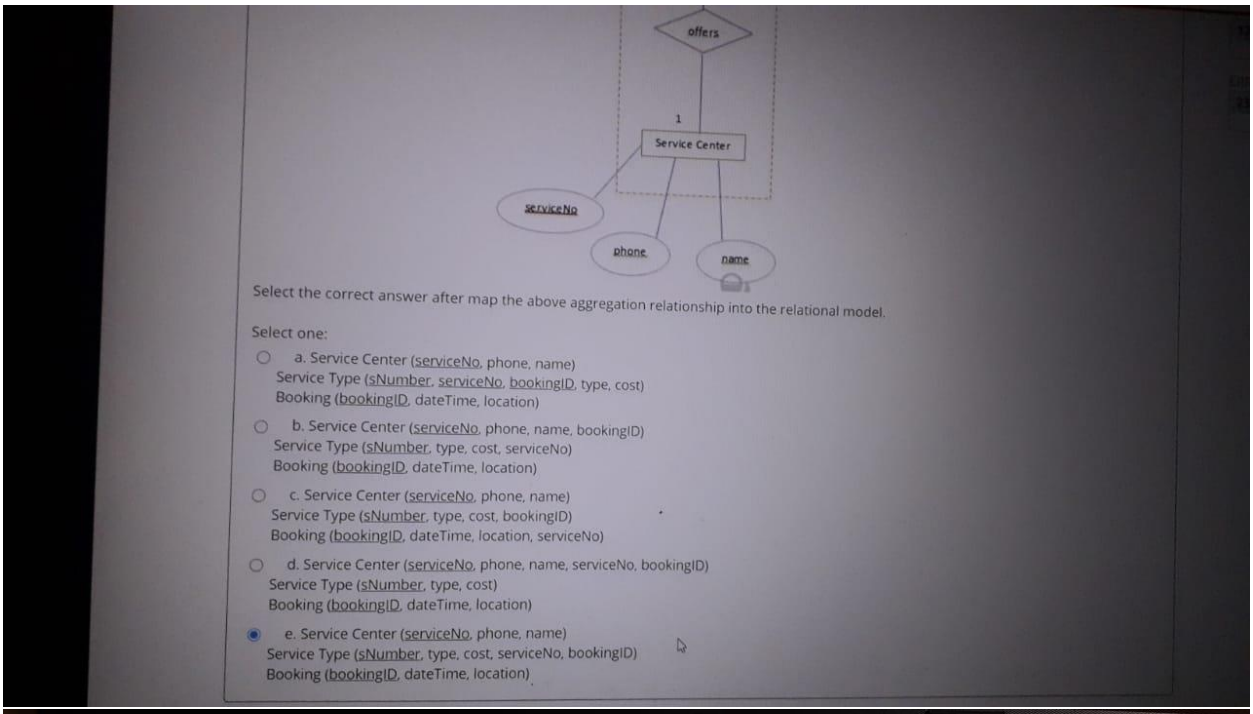
Consider the EER diagram given below:



Which of the following is true related to the above diagram?

Select one or more:

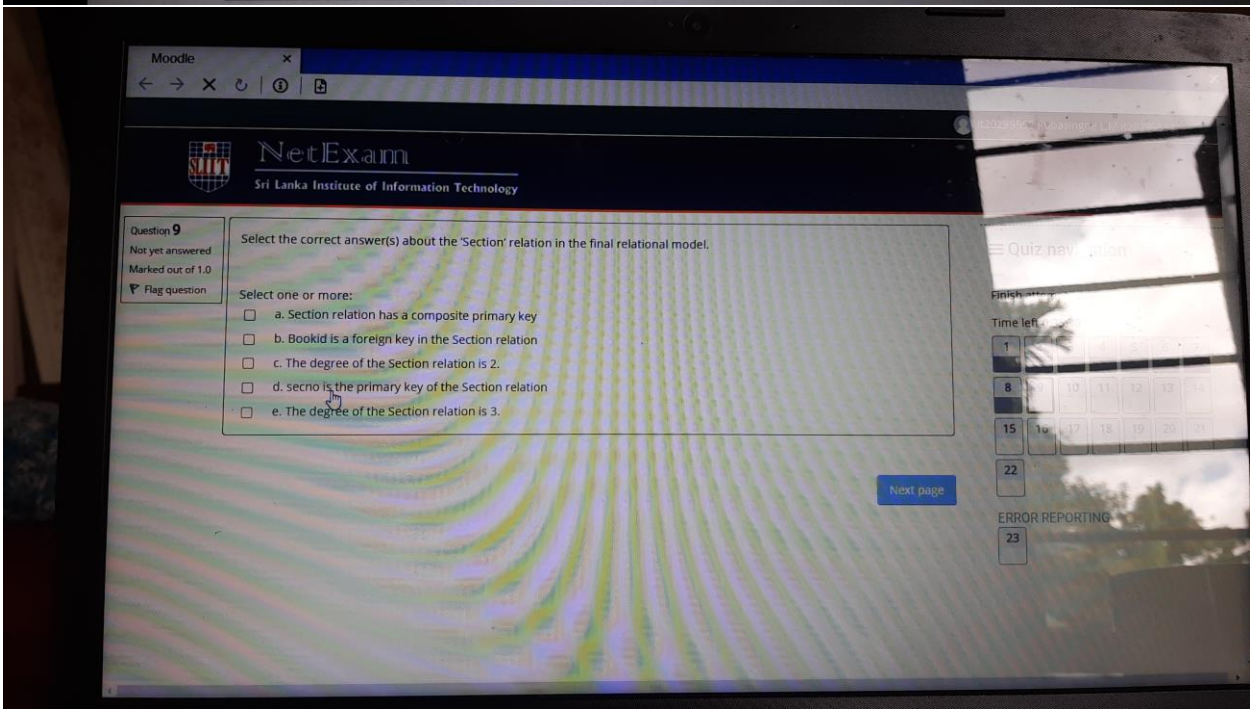
- [ ] a. There can be many batches with same value for batch
- [ ] b. Multiple lecturers can teach same module to the same group
- [x] c. A group cannot exist without a batch
- [x] d. There can be a lecturer who teach one module to more than one group
- [ ] e. There can be many groups with the same group ID in the system



Select the correct answer after map the above aggregation relationship into the relational model.

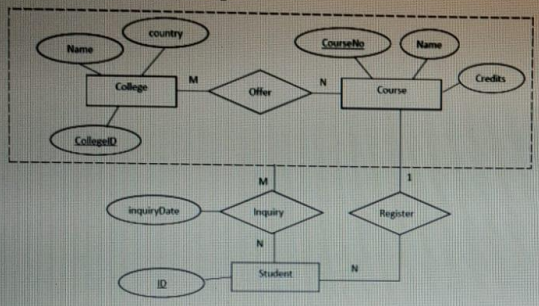
Select one:

- a. Service Center (serviceNo, phone, name)
  Service Type (sNumber, serviceNo, bookingID, type, cost)
  Booking (bookingID, dateTime, location)
- b. Service Center (serviceNo, phone, name, bookingID)
  Service Type (sNumber, type, cost, serviceNo)
  Booking (bookingID, dateTime, location)
- c. Service Center (serviceNo, phone, name)
  Service Type (sNumber, type, cost, bookingID)
  Booking (bookingID, dateTime, location, serviceNo)
- d. Service Center (serviceNo, phone, name, serviceNo, bookingID)
  Service Type (sNumber, type, cost)
  Booking (bookingID, dateTime, location)
- e. Service Center (serviceNo, phone, name)
  Service Type (sNumber, type, cost, serviceNo, bookingID)
  Booking (bookingID, dateTime, location)



Question 6
Not yet answered
Marked out of 1.0
Flag question

Consider the following EER diagram:



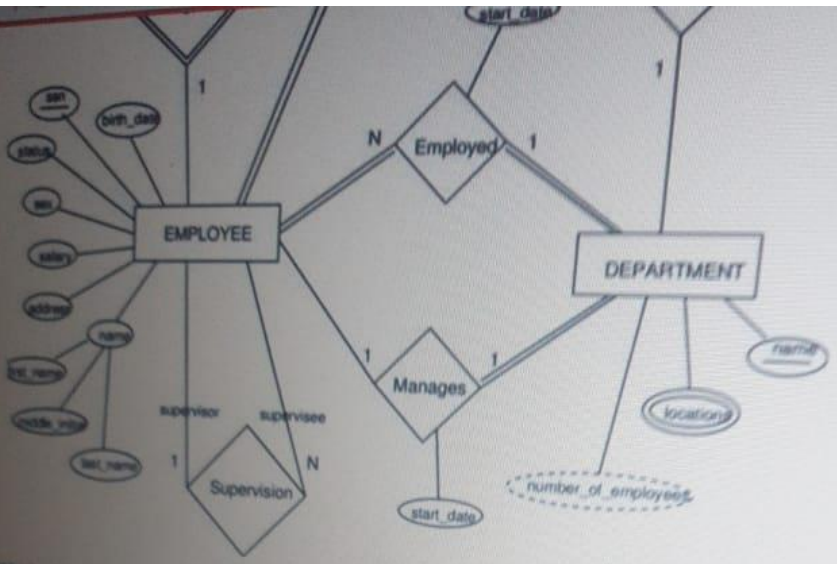
Which of the following is true related to the above diagram?

Select one or more:

- [x] a. Each course in the system has an unique course no
- [ ] b. There can be courses with no students
- [ ] c. Even though student may inquire about many course offerings he might not register to any course
- [x] d. Students may inquire about multiple courses offered by colleges
- [ ] e. Each course has a course number which is unique for each college

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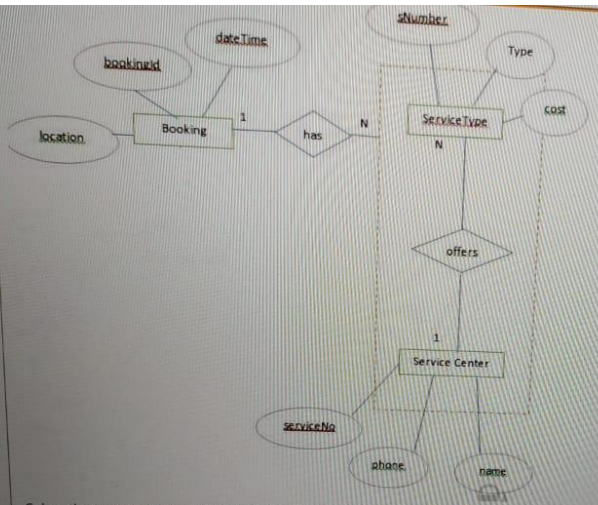


Which of the followings are correct with respect to the above diagram.

Select one or more:

- a. There could be Employees who had not worked on any project yet.
- b. There could be Projects which had not assigned any employees yet.
- c. All departments must be managed by an employee.
- d. Name is the primary key of dependent entity
- e. There can be 1 department with multiple locations.

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ERROR REPORTING

23

Select the correct answer after map the above aggregation relationship into the relational model.

Select one:

- a. Service Center (serviceNo, phone, name, serviceNo, bookingID)
  Service Type (sNumber, type, cost)
  Booking (bookingID, dateTime, location)
- b. Service Center (serviceNo, phone, name)
  Service Type (sNumber, type, cost, bookingID)
  Booking (bookingID, dateTime, location, serviceNo)
- c. Service Center (serviceNo, phone, name)
  Service Type (sNumber, serviceNo, bookingID, type, cost)

Consider the following requirements or a construction company:

Company have multiple construction sites. Each site has a unique site number, address and construction type such as('house', 'apartment', 'shopping complex'). For each site the company estimates the number the amounts required from each raw-material and these values are stored(ex: cement packs, sand, & etc.). Raw-materials have three different types such as wall-construction materials, wiring materials, Roofing materials and timber materials. Each material is identified by a unique ID and has a unit of measurement such as liters and kilograms. There are many suppliers providing raw materials for construction companies. The suppliers have are identified by a unique supplier id and each supplier has a name, address and a phone number. These are tracked by the construction company. However, during the construction company may purchase different amount of raw-materials for a site from different suppliers at different prices.

Which of the following are correct related to the EER diagram drawn for the above description.

Select one or more:

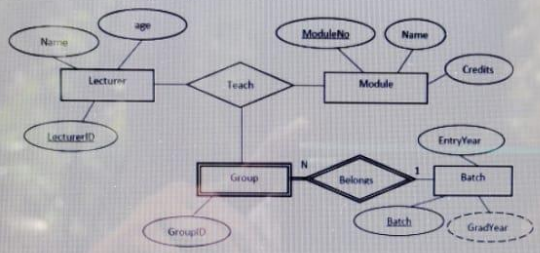
- [ ] a. There is a ternary relationship in the diagram
- [ ] b. There are only binary relationships in the diagram
- [ ] c. There is a binary relationship between site and material with a descriptive attribute
- [ ] d. Raw material types could be represented by sub classes
- [ ] e. Company is a strong entity in the EER

Next page

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Question 6
Not yet answered
Marked out of 1.0
Flag question

Consiuer the EER diagram given below:



Which of the following is true related to the above diagram?

Select one or more:

- a. There can be many groups with the same group ID in the system
- b. There can be a lecturer who teach one module to more than one group
- c. There can be many batches with same value for batch
- d. Multiple lecturers can teach same module to the same group
- e. A group cannot exist without a batch

Next page

Question 5
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out of 1.0
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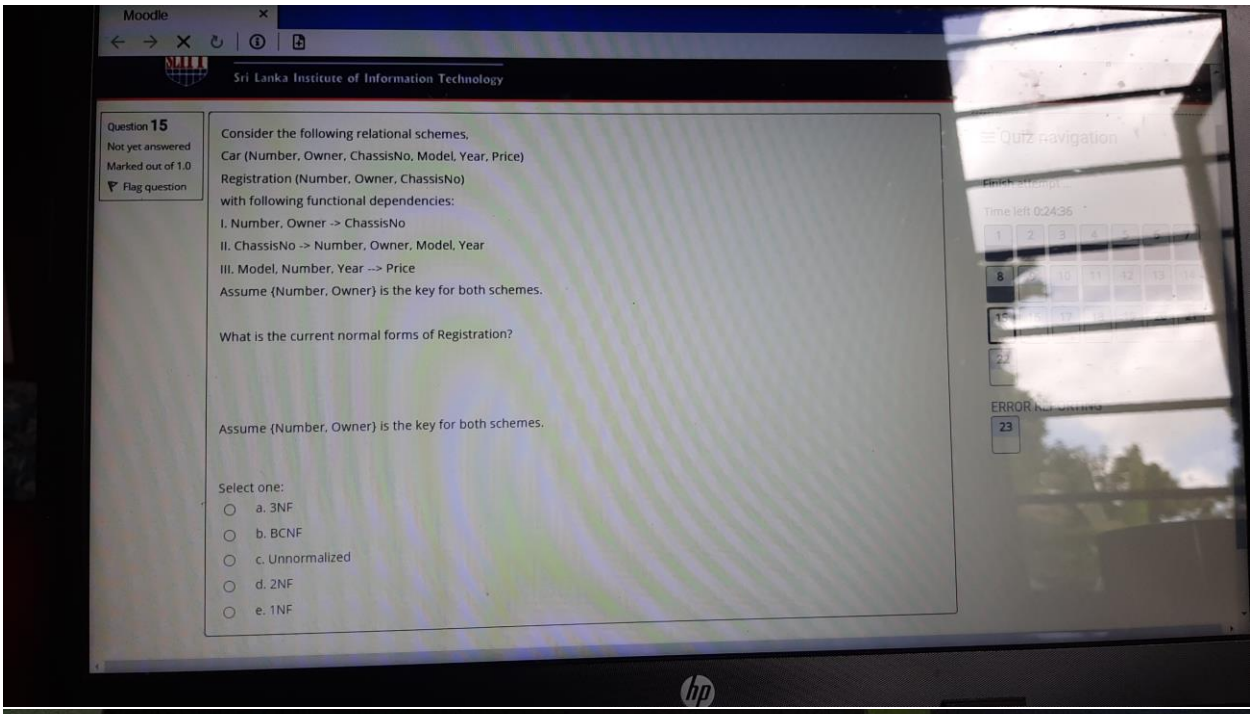
Consider the following description:

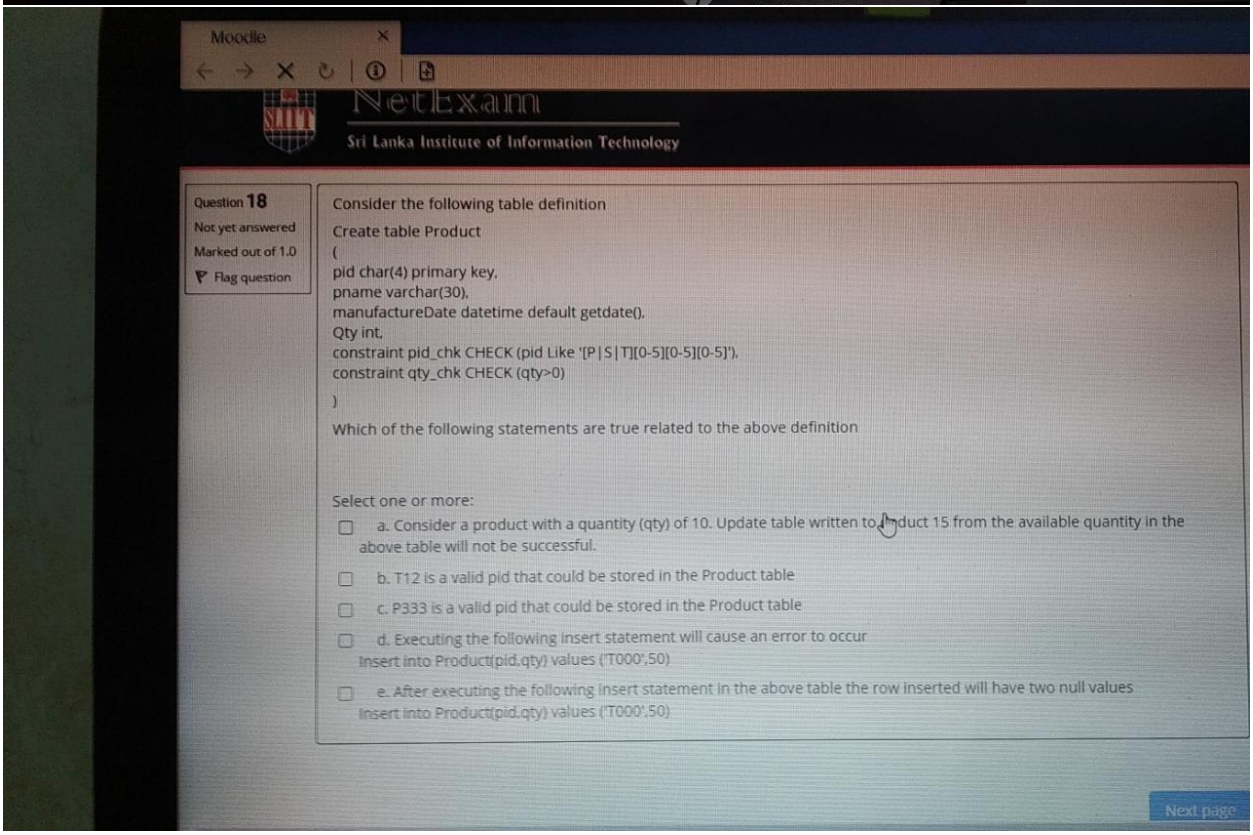
A university has two types of rooms namely lecture halls and laboratories. Lecture halls have a capacity and a number of resources such as whiteboard, podium and projector. Laboratory classes also have a capacity and number of resources. These are located in different buildings in the campus known by names such as 'Block A','Block B' and 'Block C'. Each room has a number unique to each building. There are batches taken to the universities. They are identified by the intake year and intake name (for ex: 2021 Regular intake). A batch may have several groups such as G1, G2, G3 & etc. Each group has number of students and group name is unique within each batch. During time tabling, a rooms are allocated for groups to conduct classes related to. The class name(such as 'lecture' and 'tutorial'), start time and the end time where the room will be ha should be recorded.

Which of the following are true related to the above :

Select one or more:

- a. If building is an entity room will be a weak entity
- b. Resources could be tracked using a multi-valued attribute
- c. There are only strong entities in this description
- d. The type of the rooms could be represented using sub classes
- e. The type of the rooms could be represented as an attribute


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Question 15
Not yet answered
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Flag question
Consider the following relational schemes.
Car (Number, Owner, ChassisNo, Model, Year, Price)
Registration (Number, Owner, ChassisNo)
with following functional dependencies:
I. Number, Owner -> ChassisNo
II. ChassisNo -> Number, Owner, Model, Year
III. Model, Number, Year --> Price
Assume {Number, Owner} is the key for both schemes.
What is the current normal forms of Registration?
Assume {Number, Owner} is the key for both schemes.
Select one:
a. 3NF
b. BCNF
c. Unnormalized
d. 2NF
e. 1NF
Quiz navigation
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NetExam
Sri Lanka Institute of Information Technology
Question 18
Not yet answered
Marked out of 1.0
Flag question
Consider the following table definition
Create table Product
(
pid char(4) primary key,
pname varchar(30),
manufactureDate datetime default getdate(),
Qty int,
constraint pid_chk CHECK (pid Like '[P|S|T][0-5][0-5][0-5]'),
constraint qty_chk CHECK (qty>0)
)
Which of the following statements are true related to the above definition
Select one or more:
a. Consider a product with a quantity (qty) of 10. Update table written to reduct 15 from the available quantity in the above table will not be successful.
b. T12 is a valid pid that could be stored in the Product table
c. P333 is a valid pid that could be stored in the Product table
d. Executing the following insert statement will cause an error to occur
Insert into Product(pid,qty) values ('T000',50)
e. After executing the following insert statement in the above table the row inserted will have two null values
Insert into Product(pid,qty) values ('T000',50)
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Question 18
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Flag question

Consider the following table definition

```
Create table Product
(
pid char(4) primary key,
pname varchar(30),
manufactureDate datetime default getdate(),
Qty int,
constraint pid_chk CHECK (pid Like '[P|S|T][0-5][0-5][0-5]'),
constraint qty_chk CHECK (qty>0)
)
```

Which of the following statements are true related to the above definition

Select one or more:

- [ ] a. Consider a product with a quantity (qty) of 10. Update table written to deduct 15 from the available quantity in the above table will not be successful.
- [ ] b. T12 is a valid pid that could be stored in the Product table
- [ ] c. P333 is a valid pid that could be stored in the Product table
- [ ] d. Executing the following insert statement will cause an error to occur
  Insert into Product(pid,qty) values ('T000',50)
- [ ] e. After executing the following insert statement in the above table the row inserted will have two null values
  Insert into Product(pid,qty) values ('T000',50)

Next page

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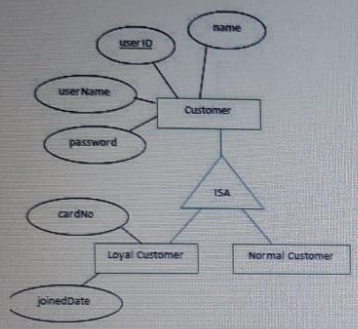
Time left 0:28:20

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ERROR REPORTING
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11
answered
out of 1.0
question

Consider the following EER diagram.



Which of the following statements are correct related to mapping the above EER model to the relational model.

Select one or more:

- [ ] a. Option 2 is more suitable
- [x] b. Option 1 would have created relations for Customer, Loyal and Normal
- [ ] c. Option 2 is not suitable
- [x] d. Option 3 and 4 will result in null values
- [ ] e. Option 3 and 4 would have created relation for Customer

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Finish attempt

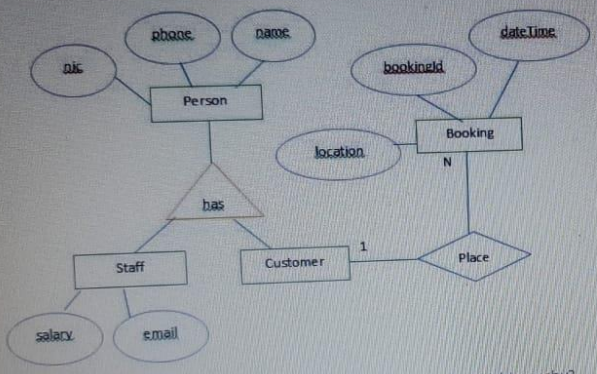
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ERROR REPORT
23

12
answered
out of 1.0
question

Consider the following EER diagram. Note that Staff and Customer cover Person.



Which option below is the most appropriate mapping for the ISA hierarchy?

Select one:

- a. Option 2
- b. Option 3
- c. Option 1 and Option 4
- d. Option 4
- e. Option 1

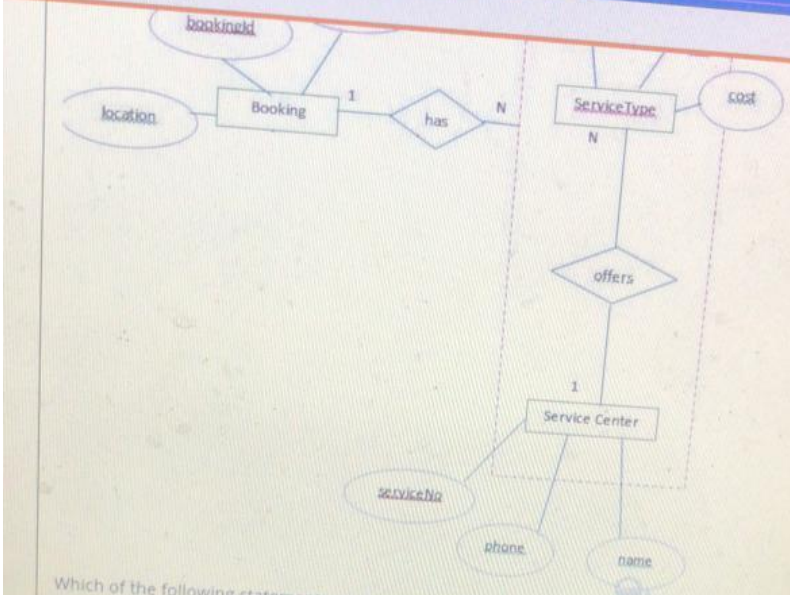
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ERROR REPORTING
23



Which of the following statements are incorrect related to mapping the above ER model to the relational model.

Select one or more:

- a. serviceNo is a foreign key in the serviceType relation
- b. bookingID is a foreign key in the Offers relation
- c. bookingID is a foreign key in the serviceType relation
- d. sNumber is a foreign key in the Service Center relation
- e. bookingID is a foreign key in the service Center relation

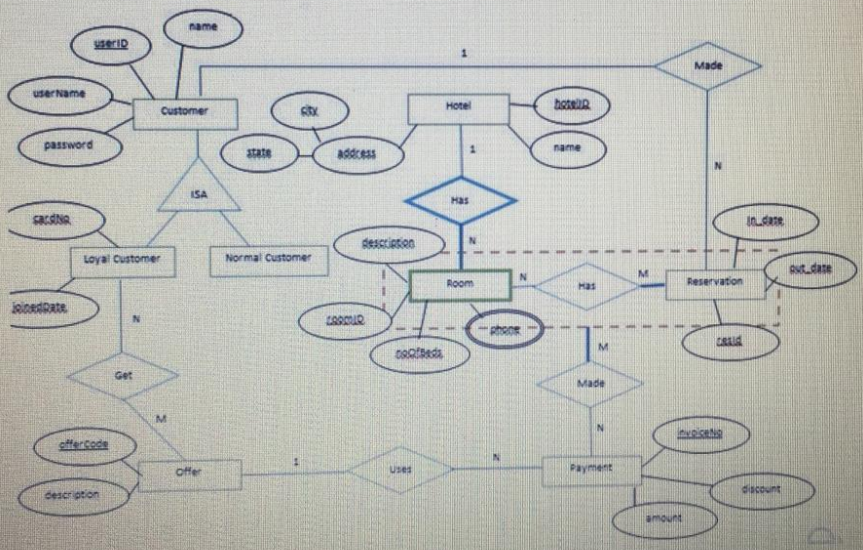
Question **10**

Not yet answered

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Flag question

Which answer gives the number of tables in the final relational model?



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Question **13**

Not yet answered

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Flag question

Consider the following relational scheme with all atomic values.

Academic_Staff (SID, FacultyID, FacultyLocation, FacultyPhone, StaffName, StaffPosition, HoursPerWeek) with following functional dependencies.

SID -> StaffName, StaffPosition, FacultyID, FacultyLocation, FacultyPhone

FacultyID -> FacultyLocation, FacultyPhone

FacultyLocation -> FacultyID, FacultyPhone

FacultyPhone -> FacultyID, FacultyLocation

What is the primary key for the relation?

Select one:

- a. StaffName
- b. SID, FacultyID
- c. FacultyPhone
- d. SID
- e. FacultyLocation

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rked out of 1.0
Flag question

Consider the following ISA hierarchy.



Which of the following statements are incorrect related to mapping the above hierarchy to the relational model.

Select one or more:

- a. Option 4 is not suitable
- b. Option 2 is more suitable
- c. Option 1 would have created relations for Person. Staff and Customer
- d. Option 3 and 4 would have created relation for Person
- e. Option 3 and 4 will result in null values

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Quiz nav

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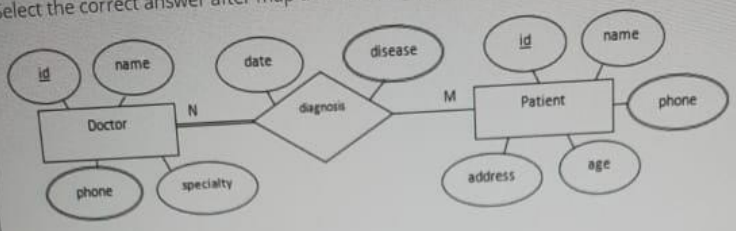
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ERROR REPORT

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Select the correct answer after map the following binary relationship into the relational model.



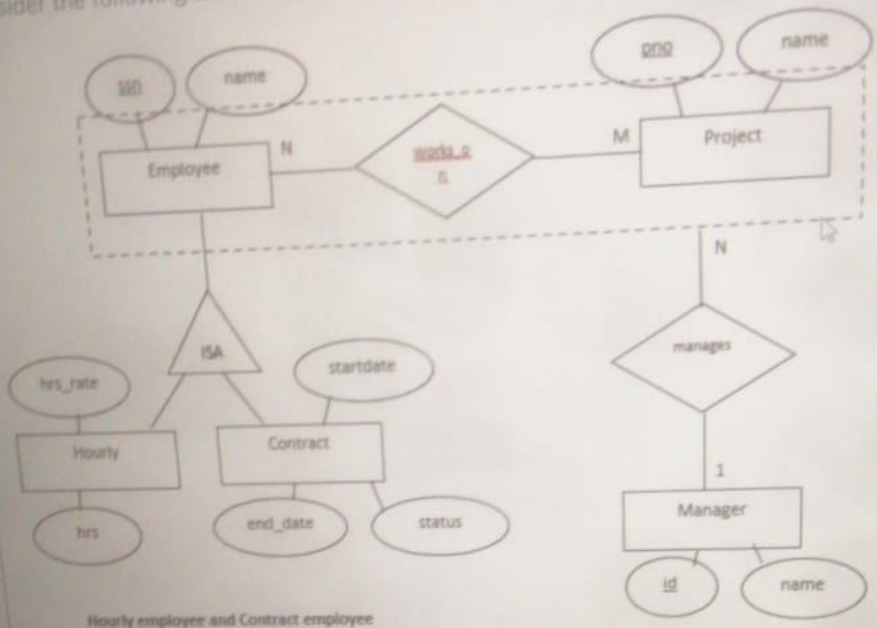
Select one:

- a. Doctor (id, name, specialty)
  Doctor_phone (id, phone)
  Patient (id, name, address, age)
  Patient_phone (id, phone)
  Diagnosis (pid, did, date, disease)
- b. Doctor (id, name, specialty, phone)
  Patient (id, name, address, age, phone)
  Diagnosis (pid, did, date, disease)
- c. Doctor (id, name, specialty)
  Doctor_phone (id, phone)
  Patient (id, name, address, age)
  Patient_phone (id, phone)
  Diagnosis (pid, did, date)
  Diagnosis_disease (pid, did, disease)
- d. Doctor (id, name, specialty)
  Doctor_phone (id, phone)

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Consider the following EER model.

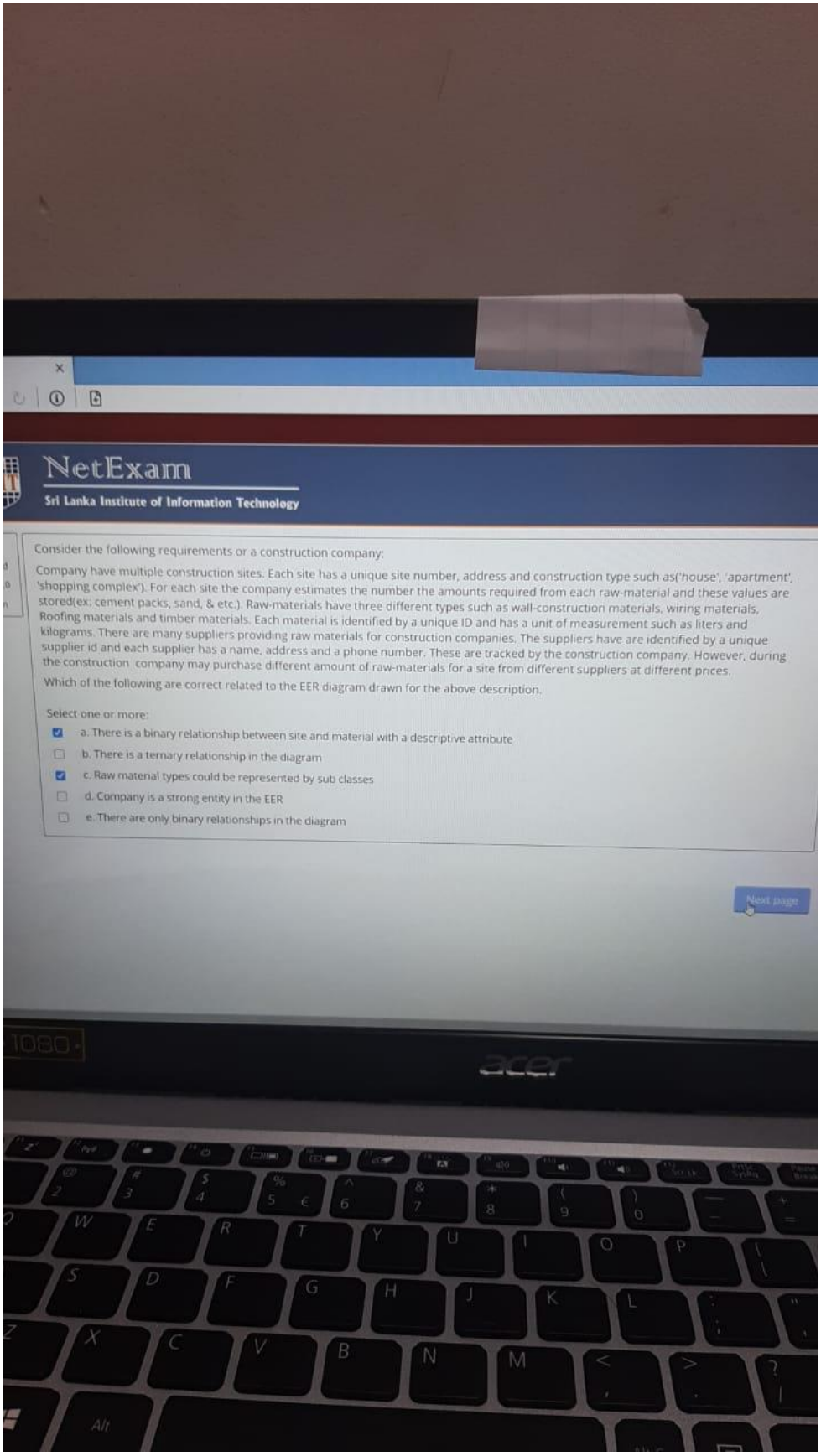


Hourly employee and Contract employee cover Employee

Which answer gives the correct number of tables in the final relational model?

Select one:

- a. 5
- b. 7


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Consider the following requirements or a construction company:
Company have multiple construction sites. Each site has a unique site number, address and construction type such as('house', 'apartment', 'shopping complex'). For each site the company estimates the number the amounts required from each raw-material and these values are stored(ex: cement packs, sand, & etc.). Raw-materials have three different types such as wall-construction materials, wiring materials, Roofing materials and timber materials. Each material is identified by a unique ID and has a unit of measurement such as liters and kilograms. There are many suppliers providing raw materials for construction companies. The suppliers have are identified by a unique supplier id and each supplier has a name, address and a phone number. These are tracked by the construction company. However, during the construction company may purchase different amount of raw-materials for a site from different suppliers at different prices.
Which of the following are correct related to the EER diagram drawn for the above description.
Select one or more:
a. There is a binary relationship between site and material with a descriptive attribute
b. There is a ternary relationship in the diagram
c. Raw material types could be represented by sub classes
d. Company is a strong entity in the EER
e. There are only binary relationships in the diagram
Next page
acer

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Consider the appointments table given below

Appointments

Patient	Doctor	appointmentDate
Lakmal	Dr. Janaka	08-01-2020
Nishani	Dr. Sunila	10-01-2020
Bhagya	Dr. Janaka	07-01-2020

What is the output of the following SQL query?

```
SELECT Count(*)
FROM ((SELECT Patient, Doctor
        FROM Appointments) AS S
       INNER JOIN (SELECT Doctor, appointmentDate
                   FROM Appointments) AS T );
```

Select one:

a. 3

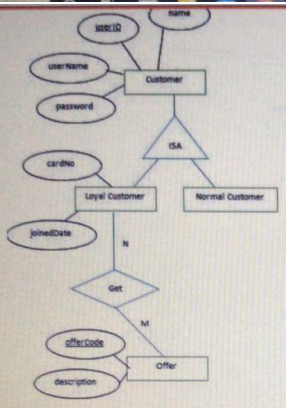
b. 6

c. 9

d. 5

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Flag question



Which option below is the most appropriate mapping for the ISA hierarchy?

Select one:

- a. Option 1
- b. Option 1 and Option 4
- c. Option 2
- d. Option 4
- e. Option 3

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13
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Consider the following relational scheme with all atomic values.

Academic_Staff (SID, FacultyID, FacultyLocation, FacultyPhone, StaffName, StaffPosition, HoursPerWeek) with following functional dependencies.

SID -> StaffName, StaffPosition, FacultyID, FacultyLocation, FacultyPhone

FacultyID -> FacultyLocation, FacultyPhone

FacultyLocation -> FacultyID, FacultyPhone

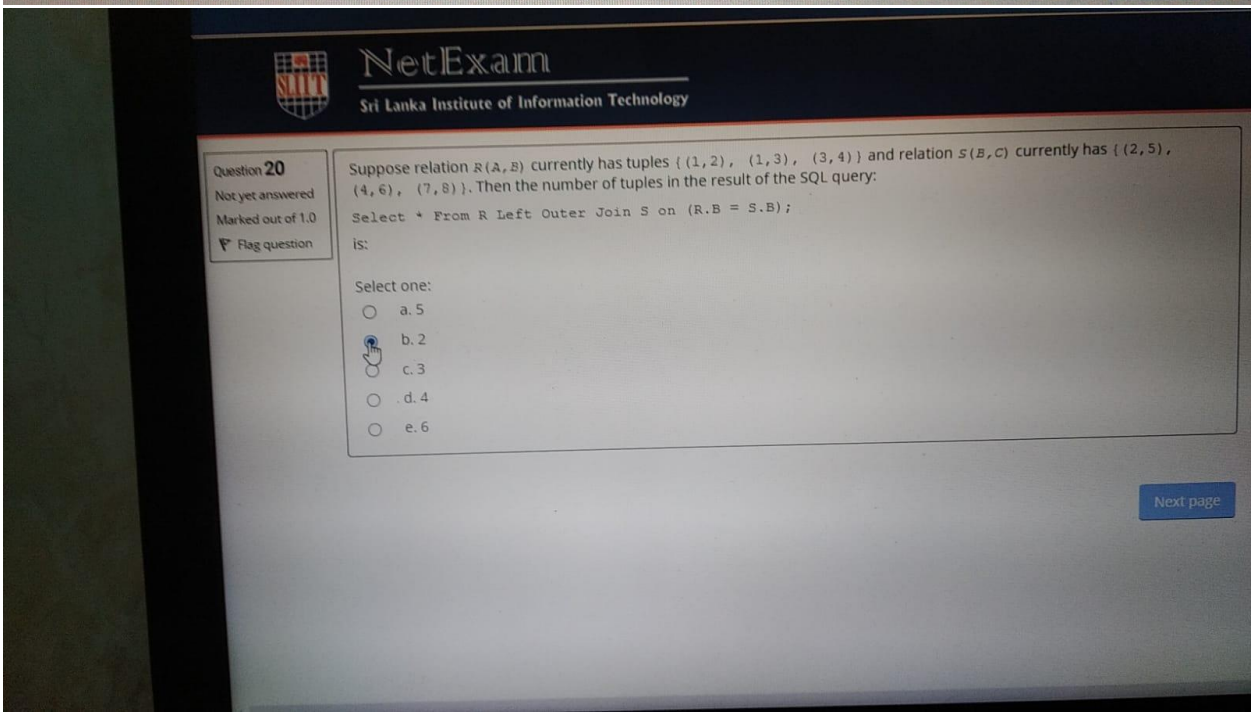
FacultyPhone -> FacultyID, FacultyLocation

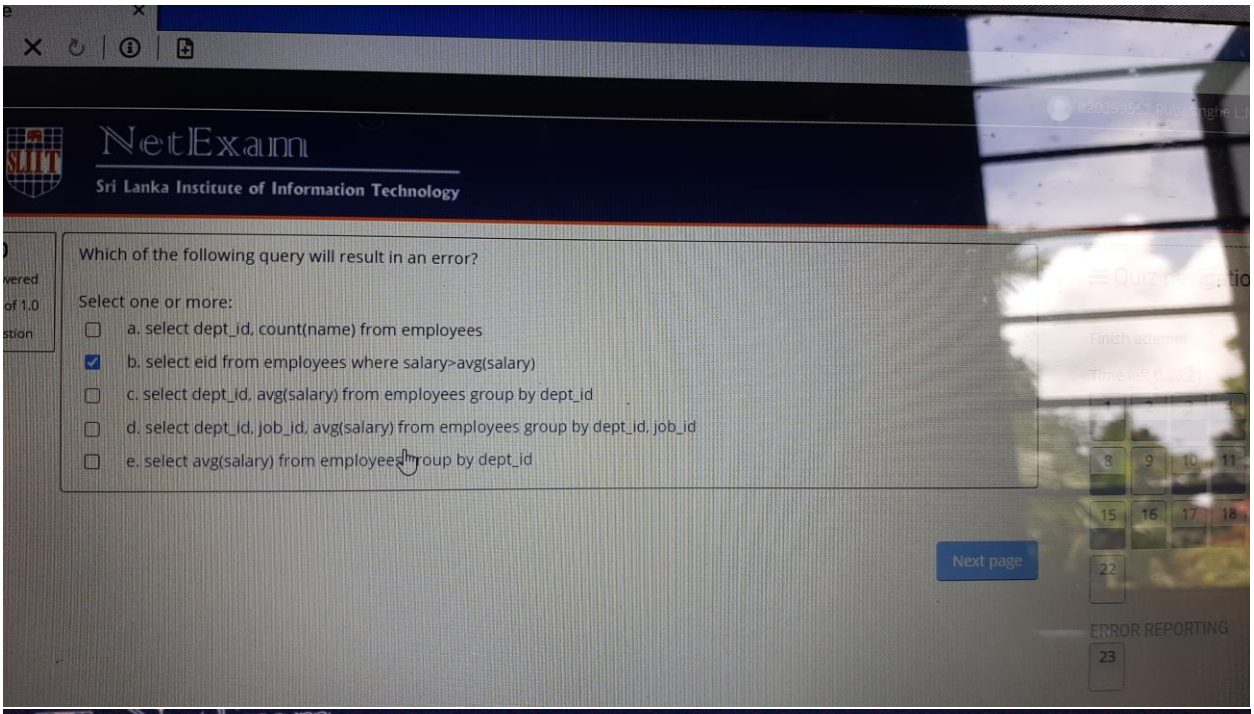
What is the primary key for the relation?

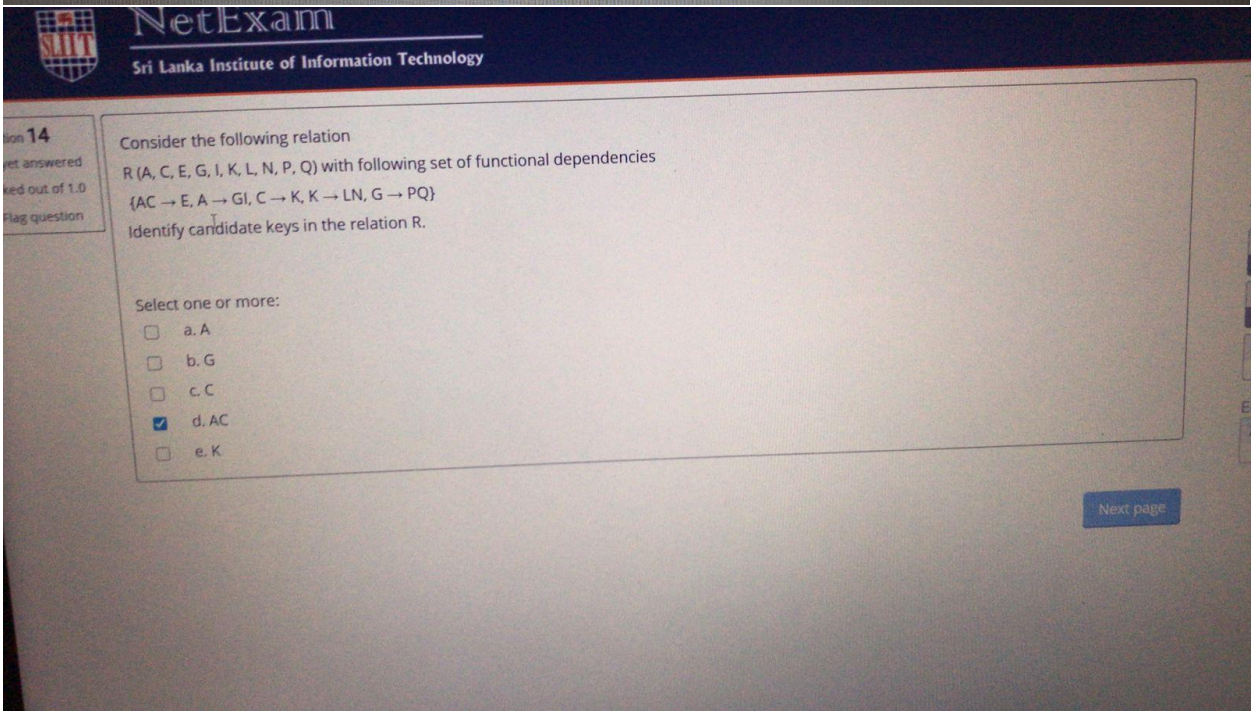
Select one:

- a. FacultyLocation
- b. SID
- c. FacultyPhone
- d. StaffName
- e. SID, FacultyID

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Which of the following query will result in an error?
Select one or more:
a. select dept_id, count(name) from employees
b. select eid from employees where salary>avg(salary)
c. select dept_id, avg(salary) from employees group by dept_id
d. select dept_id, job_id, avg(salary) from employees group by dept_id, job_id
e. select avg(salary) from employees group by dept_id
Next page
ERROR REPORTING


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14
Consider the following relation
R (A, C, E, G, I, K, L, N, P, Q) with following set of functional dependencies
{AC → E, A → GI, C → K, K → LN, G → PQ}
Identify candidate keys in the relation R.
Select one or more:
a. A
b. G
c. C
d. AC
e. K
Next page

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Question 20
Not yet answered
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Flag question

Suppose relation R(A,B) currently has tuples {(1,2), (1,3), (3,4)} and relation S(B,C) currently has {(2,5), (4,6), (7,8)}. Then the number of tuples in the result of the SQL query:

```
Select * From R Left Outer Join S on (R.B = S.B);
```

is:

Select one:

- a. 5
- b. 2
- c. 3
- d. 4
- e. 6

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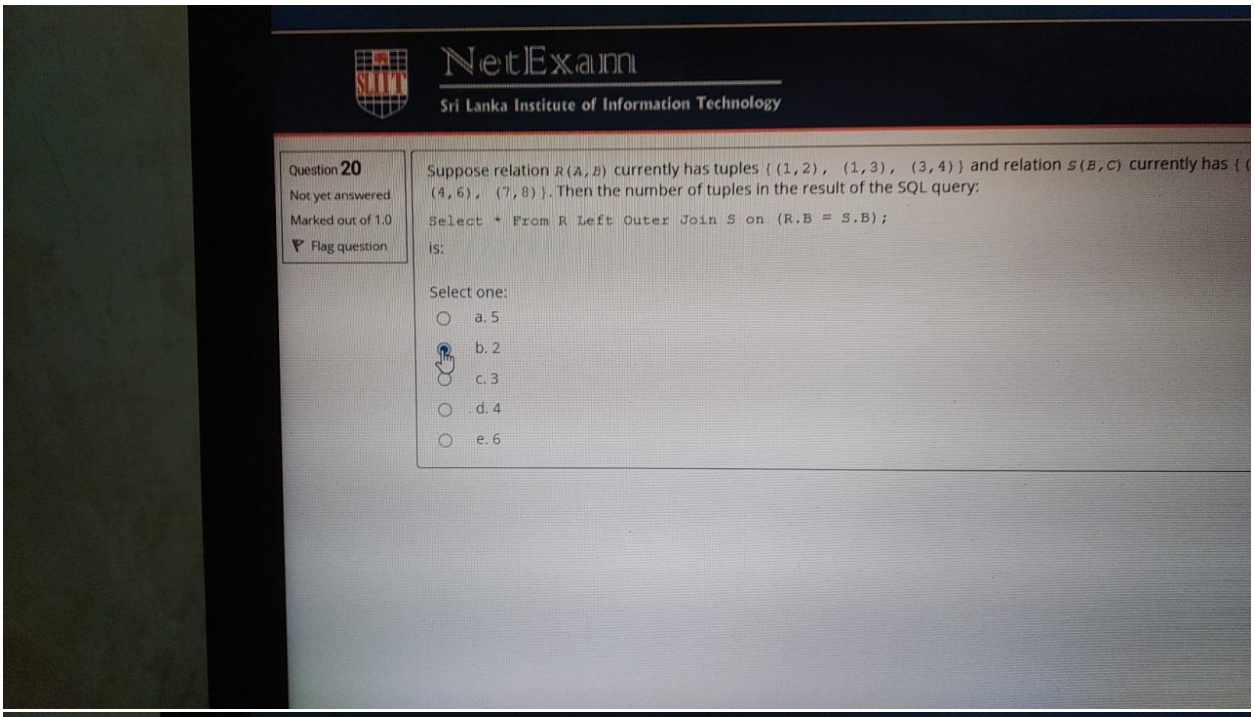
Relation R has eight attributes RUSQTP. Please consider relation R contains only atomic values.

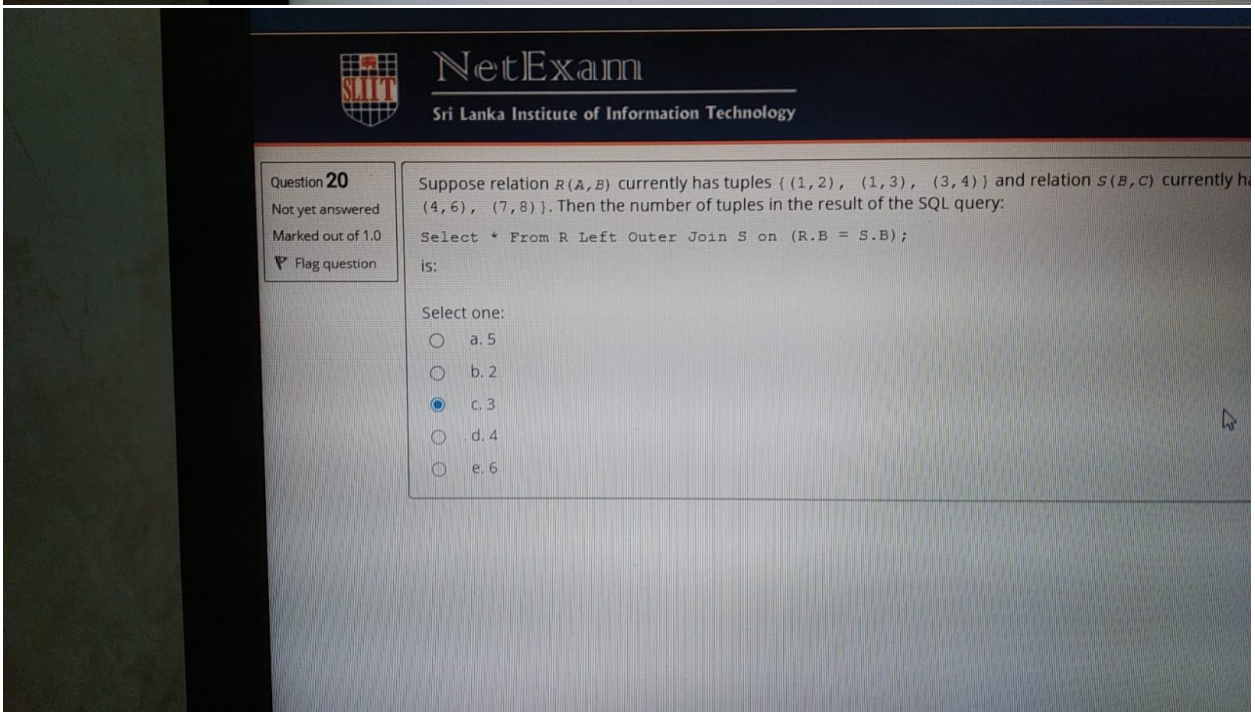
F = {R -> S, Q -> RU, T -> P} is a set of functional dependencies that hold for R.

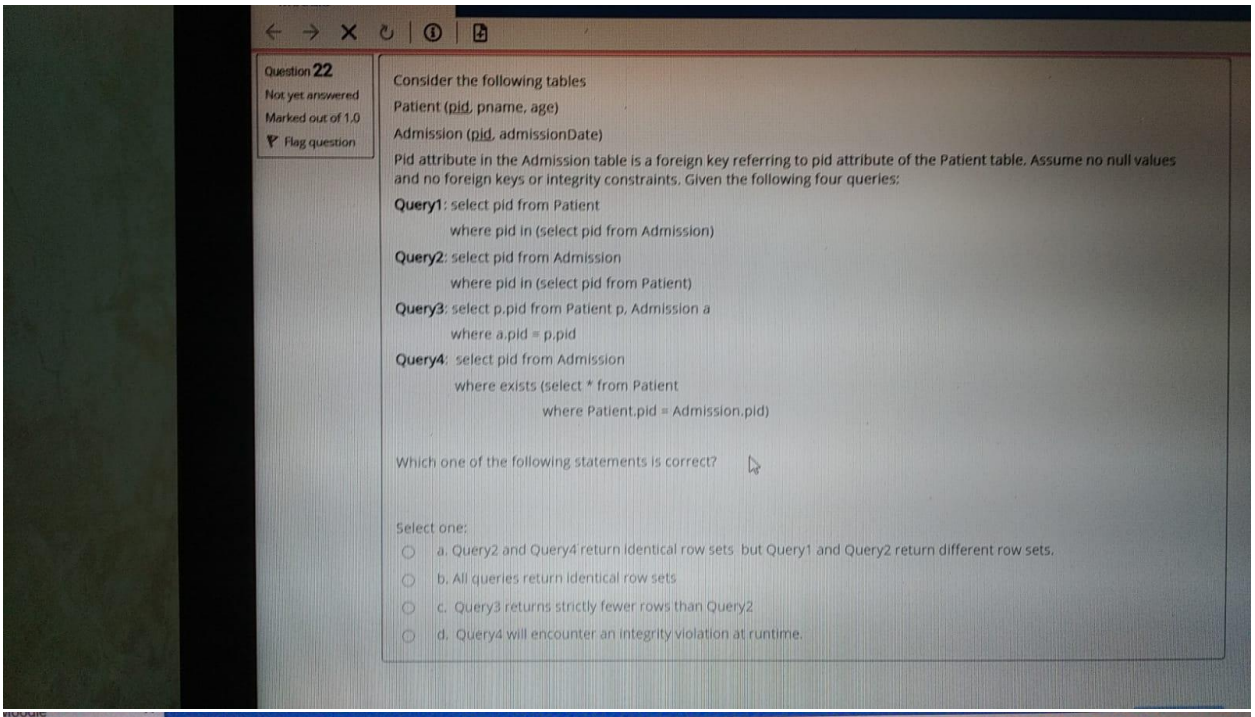
What is the normal form that the above relation is in?

Select one:

- a. Unnormalized Form
- b. BCNF
- c. 3NF
- d. 2NF
- e. 1NF


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Question 20
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Flag question
Suppose relation R(A,B) currently has tuples {(1,2), (1,3), (3,4)} and relation S(B,C) currently has {(
(4,6), (7,8)}. Then the number of tuples in the result of the SQL query:
Select * From R Left Outer Join S on (R.B = S.B);
is:
Select one:
a. 5
b. 2
c. 3
d. 4
e. 6


NetExam
Sri Lanka Institute of Information Technology
Question 20
Not yet answered
Marked out of 1.0
Flag question
Suppose relation R(A,B) currently has tuples {(1,2), (1,3), (3,4)} and relation S(B,C) currently ha
(4,6), (7,8)}. Then the number of tuples in the result of the SQL query:
Select * From R Left Outer Join S on (R.B = S.B);
is:
Select one:
a. 5
b. 2
c. 3
d. 4
e. 6



Question 22
Not yet answered
Marked out of 1.0
Flag question

Consider the following tables

Patient (pid, pname, age)

Admission (pid, admissionDate)

Pid attribute in the Admission table is a foreign key referring to pid attribute of the Patient table. Assume no null values and no foreign keys or integrity constraints. Given the following four queries:

Query1: select pid from Patient
where pid in (select pid from Admission)

Query2: select pid from Admission
where pid in (select pid from Patient)

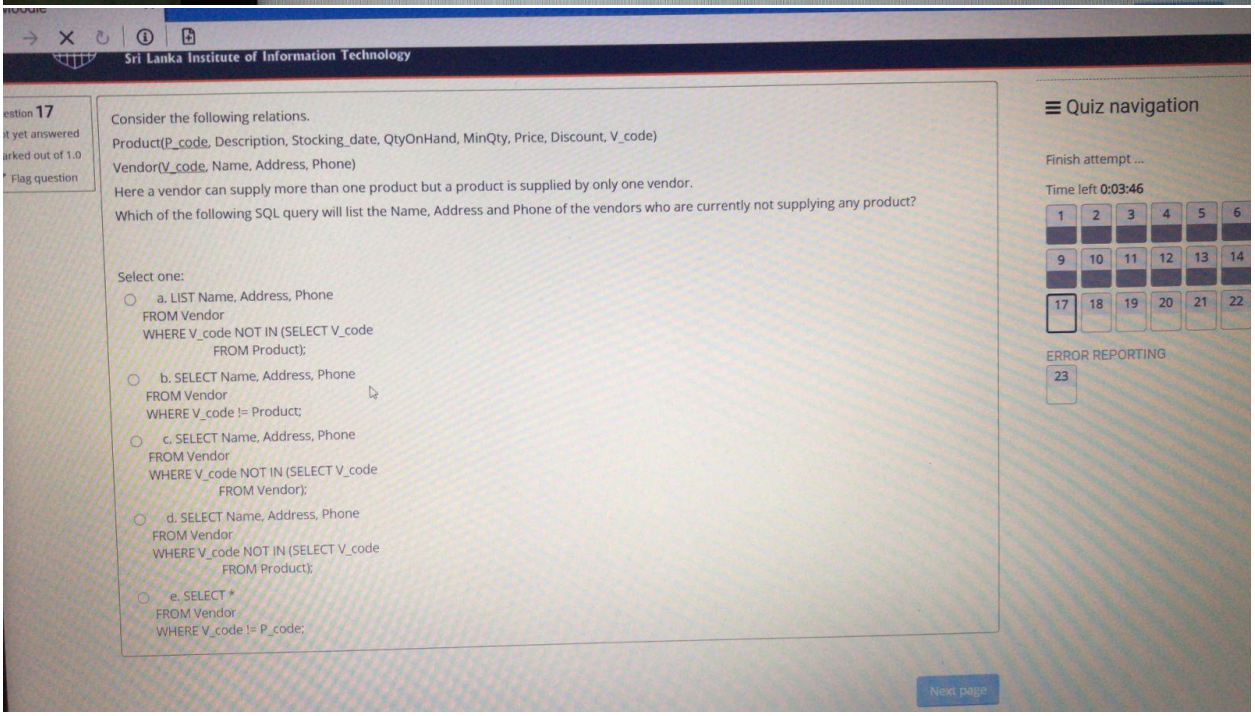
Query3: select p.pid from Patient p, Admission a
where a.pid = p.pid

Query4: select pid from Admission
where exists (select * from Patient
where Patient.pid = Admission.pid)

Which one of the following statements is correct?

Select one:

- a. Query2 and Query4 return identical row sets but Query1 and Query2 return different row sets.
- b. All queries return identical row sets
- c. Query3 returns strictly fewer rows than Query2
- d. Query4 will encounter an integrity violation at runtime.



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Consider the following relations.

Product(P_code, Description, Stocking_date, QtyOnHand, MinQty, Price, Discount, V_code)

Vendor(V_code, Name, Address, Phone)

Here a vendor can supply more than one product but a product is supplied by only one vendor.

Which of the following SQL query will list the Name, Address and Phone of the vendors who are currently not supplying any product?

Select one:

- a. LIST Name, Address, Phone
  FROM Vendor
  WHERE V_code NOT IN (SELECT V_code
  FROM Product);
- b. SELECT Name, Address, Phone
  FROM Vendor
  WHERE V_code != Product;
- c. SELECT Name, Address, Phone
  FROM Vendor
  WHERE V_code NOT IN (SELECT V_code
  FROM Vendor);
- d. SELECT Name, Address, Phone
  FROM Vendor
  WHERE V_code NOT IN (SELECT V_code
  FROM Product);
- e. SELECT *
  FROM Vendor
  WHERE V_code != P_code;

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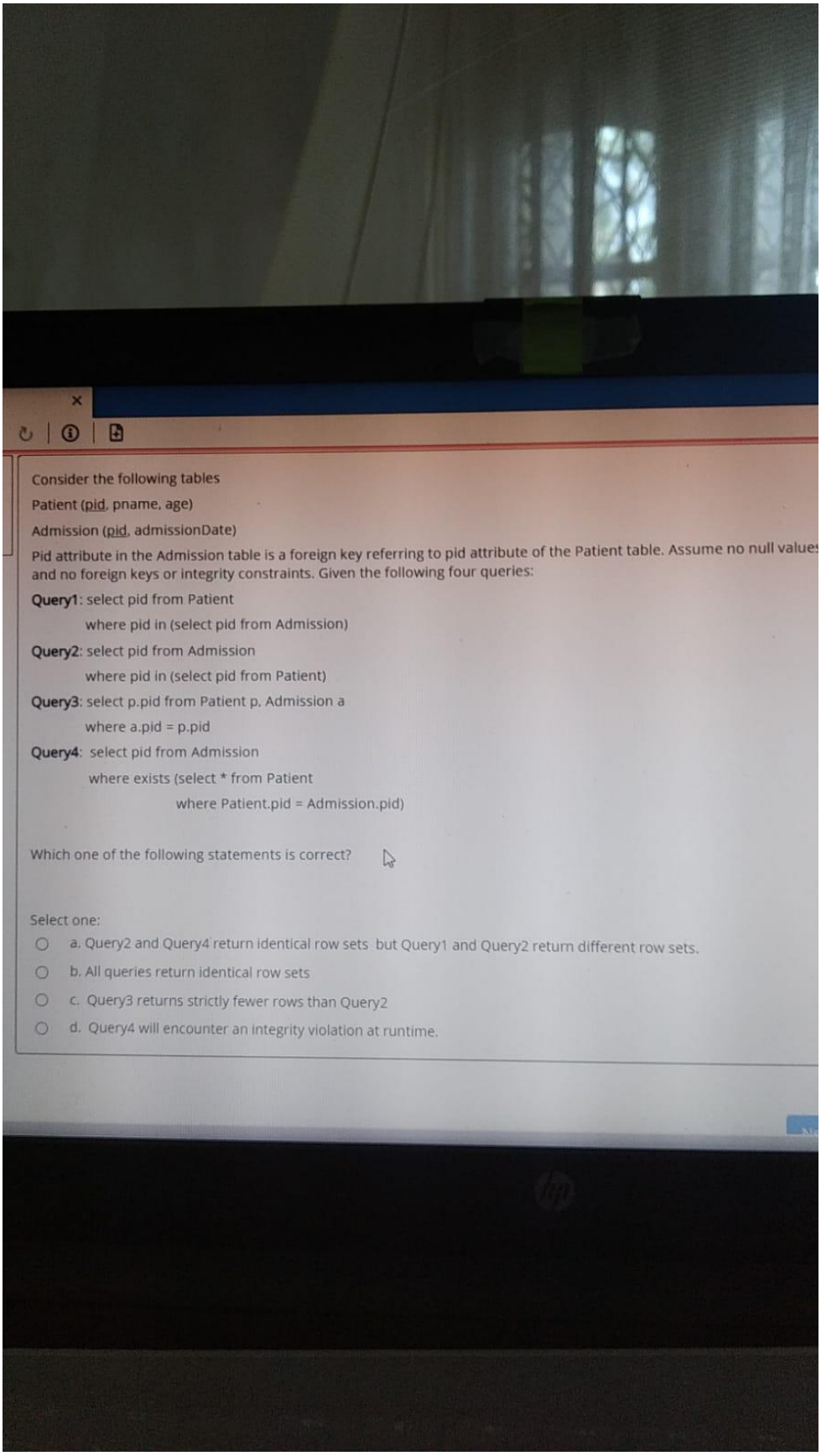
Quiz navigation

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ERROR REPORTING
23



Consider the following tables

Patient (pid, pname, age)

Admission (pid, admissionDate)

Pid attribute in the Admission table is a foreign key referring to pid attribute of the Patient table. Assume no null values and no foreign keys or integrity constraints. Given the following four queries:

Query1: select pid from Patient
where pid in (select pid from Admission)

Query2: select pid from Admission
where pid in (select pid from Patient)

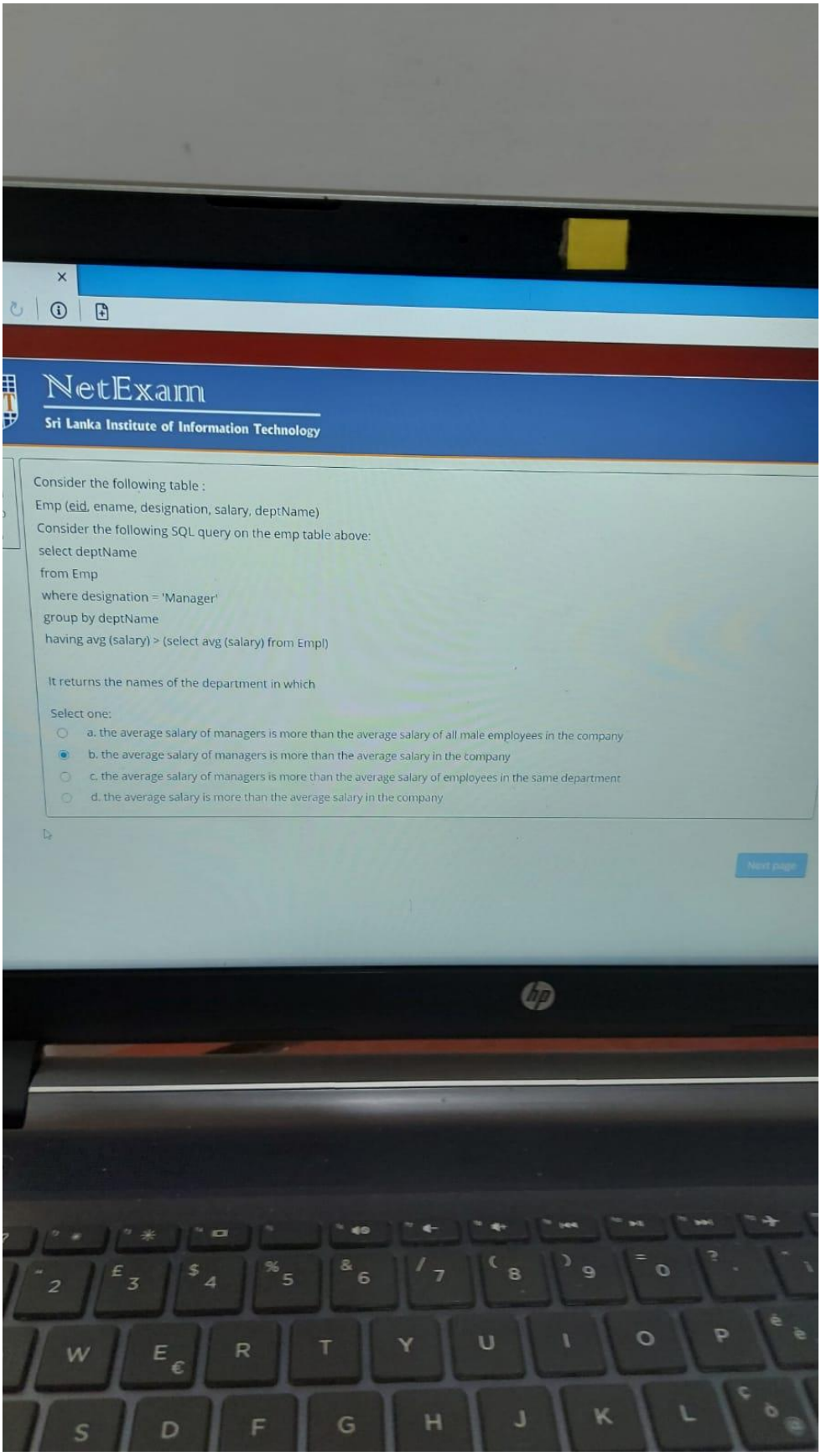
Query3: select p.pid from Patient p, Admission a
where a.pid = p.pid

Query4: select pid from Admission
where exists (select * from Patient
where Patient.pid = Admission.pid)

Which one of the following statements is correct?

Select one:

- a. Query2 and Query4 return identical row sets but Query1 and Query2 return different row sets.
- b. All queries return identical row sets
- c. Query3 returns strictly fewer rows than Query2
- d. Query4 will encounter an integrity violation at runtime.


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Consider the following table :
Emp (eid, ename, designation, salary, deptName)
Consider the following SQL query on the emp table above:
select deptName
from Emp
where designation = 'Manager'
group by deptName
having avg (salary) > (select avg (salary) from Empl)
It returns the names of the department in which
Select one:
a. the average salary of managers is more than the average salary of all male employees in the company
b. the average salary of managers is more than the average salary in the company
c. the average salary of managers is more than the average salary of employees in the same department
d. the average salary is more than the average salary in the company
Next page

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Consider a relation Lecturer as follows

Lecturer(EmpNo CHAR(03), Name VARCHAR(50), Salary REAL, Category VARCHAR(25), DateJoined DATE, DNo CHAR(02))

Consider the following details of a Lecturer.

EmpNo – 175, Name – Dulip Silva, Salary - 12,000, Category – Instructor, DateJoined - SYSDATE , DNo – 05

Which of the following SQL statements will insert the above data into Lecturer relation?

Select one or more:

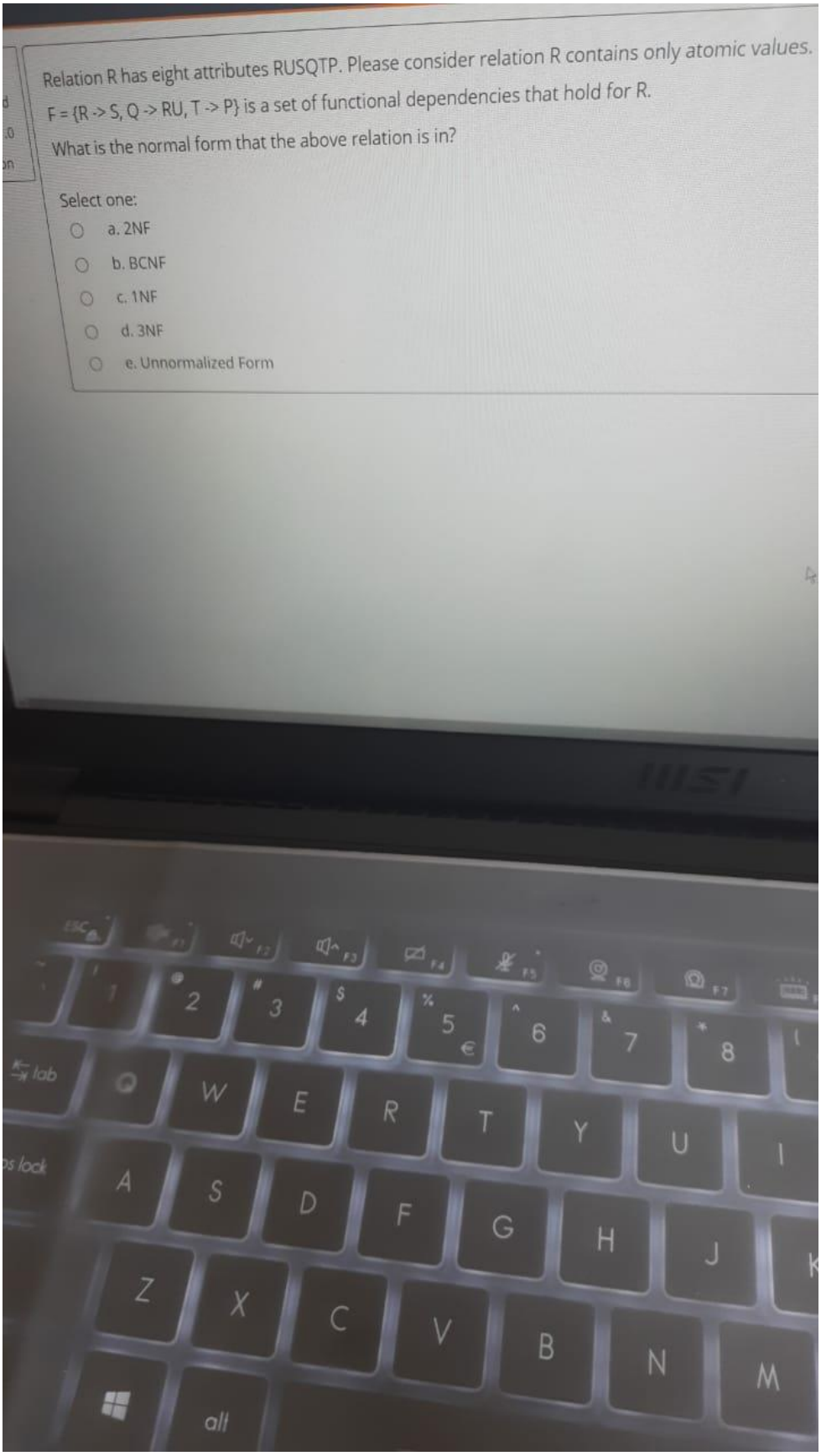
- a. INSERT (EmpNo, Name, Salary, Category , DateJoined, DNo) VALUES ('175', 'Dulip Silva', 12000, 'Instructor', SYSDATE, '05') INTO Lecturer;
- b. INSERT INTO Lecturer VALUES (175, 'Dulip Silva', 12000,'Instructor',SYSDATE, '05');
- c. INSERT INTO Lecturer (EmpNo, Name, Salary, Category ,DateJoined, DNo) VALUES (175, 'Dulip Silva', 12.000, 'Instructor', SYSDATE, '05');
- d. RT INTO Lecturer (EmpNo, Name, Salary, Category,DateJoined, DNo)VALUES ('175', 'Dulip Silva', 12000, 'Instructor', SYSDATE, '05');
- e. INSERT INTO Lecturer VALUES ('175','Dulip Silva', 12000, 'Instructor', SYSDATE, '05');

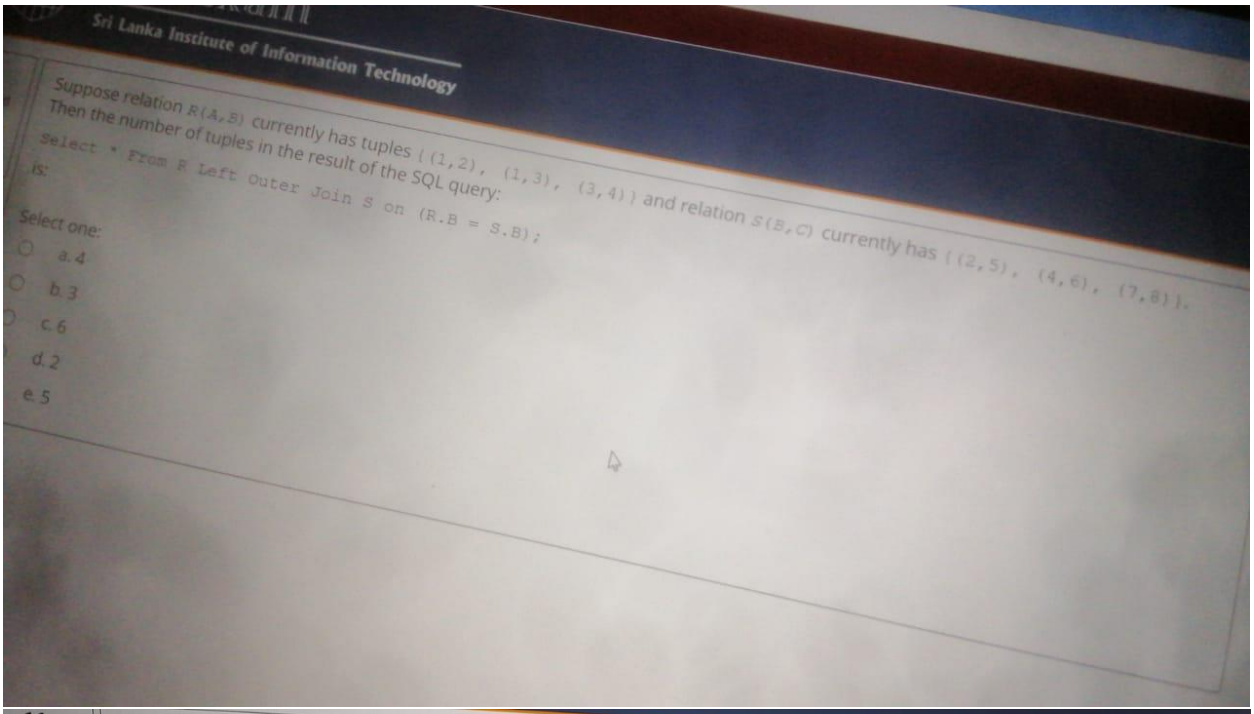
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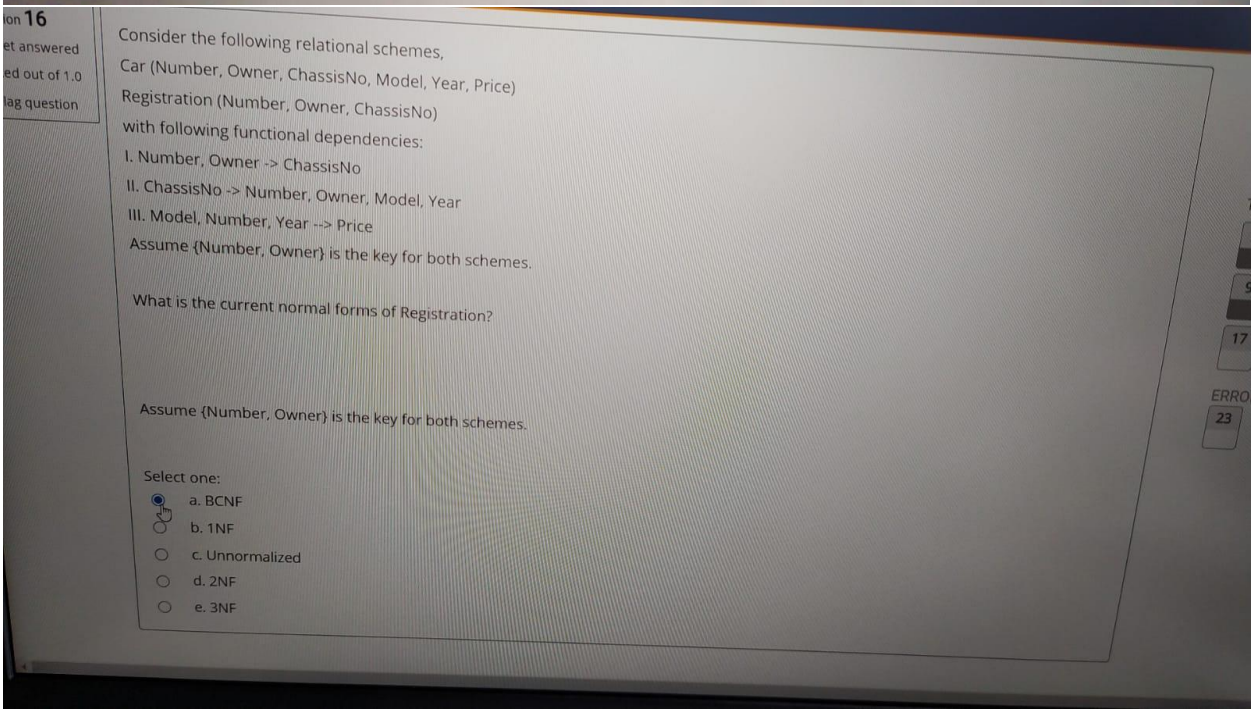
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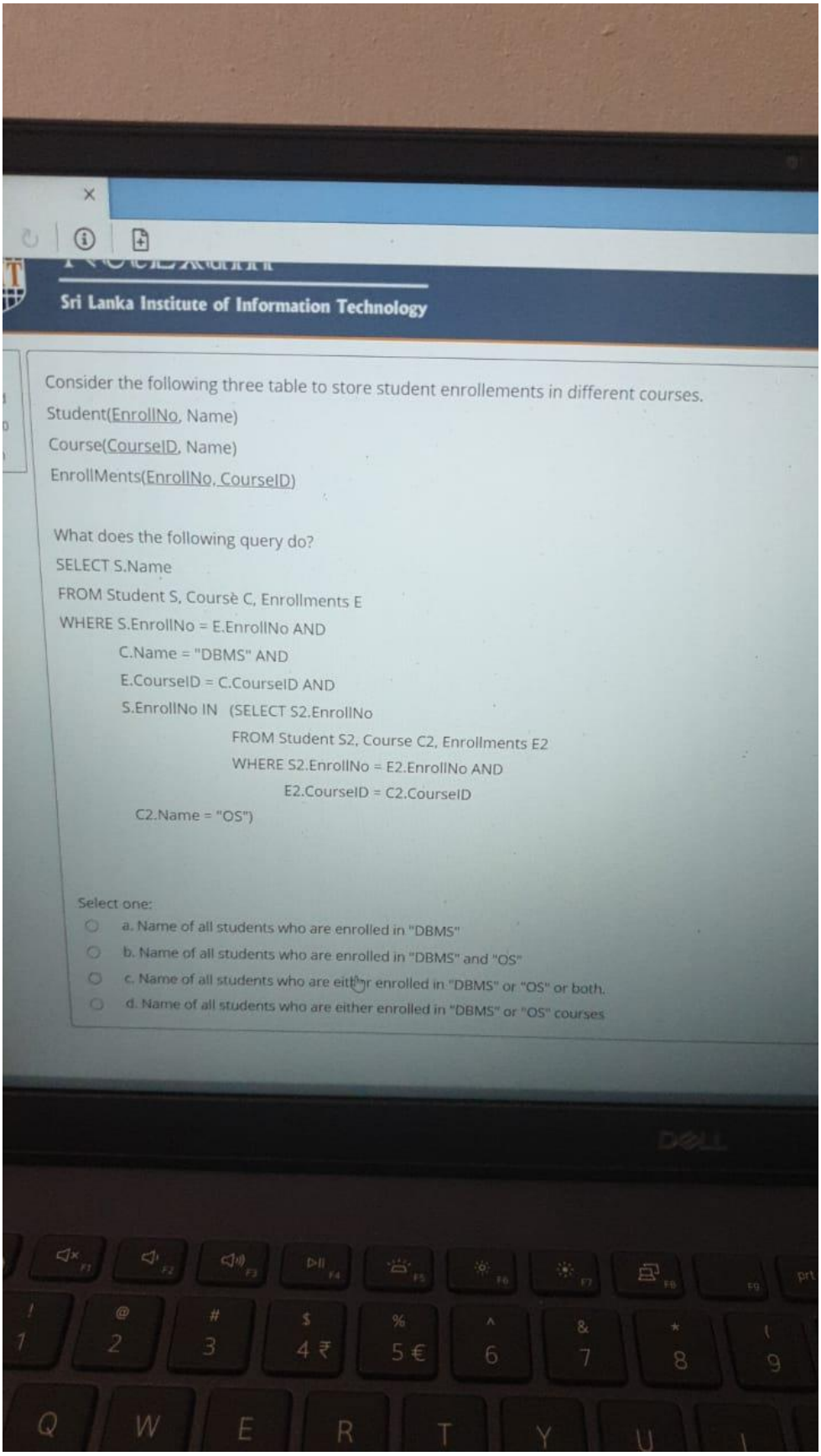
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Relation R has eight attributes RUSQTP. Please consider relation R contains only atomic values.
F = {R -> S, Q -> RU, T -> P} is a set of functional dependencies that hold for R.
What is the normal form that the above relation is in?
Select one:
a. 2NF
b. BCNF
c. 1NF
d. 3NF
e. Unnormalized Form
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Sri Lanka Institute of Information Technology
Suppose relation R(A,B) currently has tuples {(1,2), (1,3), (3,4)} and relation S(B,C) currently has {(2,5), (4,6), (7,8)}.
Then the number of tuples in the result of the SQL query:
Select * From R Left Outer Join S on (R.B = S.B);
is:
Select one:
a. 4
b. 3
c. 6
d. 2
e. 5


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lag question
Consider the following relational schemes,
Car (Number, Owner, ChassisNo, Model, Year, Price)
Registration (Number, Owner, ChassisNo)
with following functional dependencies:
I. Number, Owner -> ChassisNo
II. ChassisNo -> Number, Owner, Model, Year
III. Model, Number, Year --> Price
Assume {Number, Owner} is the key for both schemes.
What is the current normal forms of Registration?
Assume {Number, Owner} is the key for both schemes.
Select one:
a. BCNF
b. 1NF
c. Unnormalized
d. 2NF
e. 3NF
17
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Sri Lanka Institute of Information Technology
Consider the following three table to store student enrollements in different courses.
Student(EnrollNo, Name)
Course(CourseID, Name)
EnrollMents(EnrollNo, CourseID)
What does the following query do?
SELECT S.Name
FROM Student S, Coursè C, Enrollments E
WHERE S.EnrollNo = E.EnrollNo AND
C.Name = "DBMS" AND
E.CourseID = C.CourseID AND
S.EnrollNo IN (SELECT S2.EnrollNo
FROM Student S2, Course C2, Enrollments E2
WHERE S2.EnrollNo = E2.EnrollNo AND
E2.CourseID = C2.CourseID
C2.Name = "OS")
Select one:
a. Name of all students who are enrolled in "DBMS"
b. Name of all students who are enrolled in "DBMS" and "OS"
c. Name of all students who are either enrolled in "DBMS" or "OS" or both.
d. Name of all students who are either enrolled in "DBMS" or "OS" courses

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Which of the following query will result in an error?

Select one or more:

- [ ] a. select dept_id, count(name) from employees
- [ ] b. select avg(salary) from employees group by dept_id
- [ ] c. select eid from employees where salary>avg(salary)
- [ ] d. select dept_id, job_id, avg(salary) from employees group by dept_id, job_id
- [ ] e. select dept_id, avg(salary) from employees group by dept_id

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Consider the following relation

CustomerSales(CustNo, SalesDate, SalesAmount, SalesRepNo, Location)

with following set of functional dependencies,

CustNo, SalesDate –> SalesAmount, SalesRepNo, Location

SalesRepNo, SalesDate, SalesTime –> CustNo

Location, SalesDate, SalesTime –> SalesRepNo, CustNo

Identify candidate keys in the relation R.

Select one or more:

- [ ] a. CustNo,SalesRepNo)
- [ ] b. (CustNo, SalesDate)
- [ ] c. (CustNo)
- [ ] d. (SalesRepNo, SalesDate, SalesTime)
- [ ] e. (Location, SalesDate, SalesTime)

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Consider the following table:

Emp (eid, ename, designation, salary, did)

Consider the following relational query on the Emp table above:

```
SELECT salary
FROM Emp e1
WHERE 2 = (
        SELECT COUNT( DISTINCT ( e2.salary ) )
        FROM Emp e2
        WHERE e2.salary > e1.salary
    )
```

Which one of the following is the correct interpretation of the above query?

Select one:

- a. find the 2nd highest salary from table
- b. find the 3rd highest salary from table
- c. find the 4th highest salary from table
- d. find the highest salary from table

Consider the following relation

Weather (CityID,temperature,city,condition)

Which of the following query will return the names of these cities with temperature and condition whose condition is neither sunny nor cloudy.

Select one:

a. SELECT city, temperature, condition
FROM weather
WHERE condition NOT EXISTS ('sunny', 'cloudy');

b. SELECT city, temperature, condition
FROM weather
WHERE condition NOT IN ('sunny', 'cloudy')

c. SELECT city, temperature, condition
FROM weather
WHERE condition BETWEEN ('sunny', 'cloudy');

d. SELECT city, temperature, condition
FROM weather
WHERE condition IN ('sunny', 'cloudy')

e. SELECT city, temperature, condition
FROM weather
WHERE condition EXISTS ('sunny', 'cloudy')

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Consider the following schema

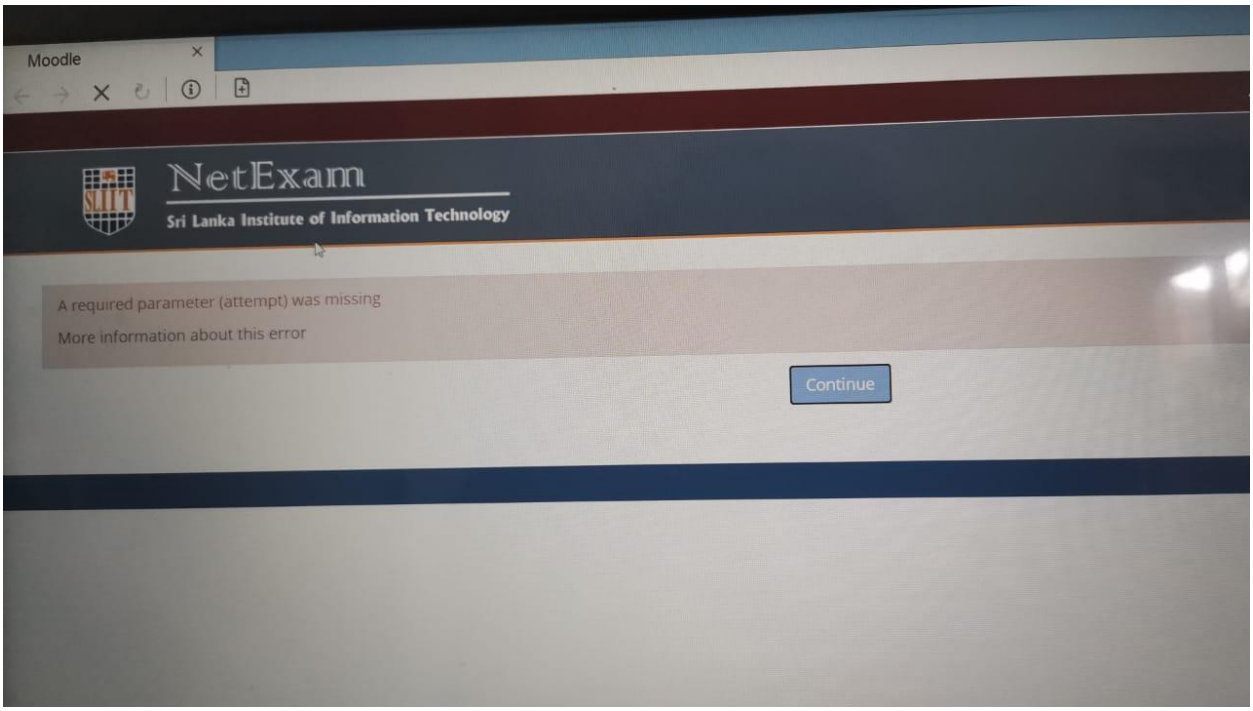
EMP(eid, ename, age, salary, did)

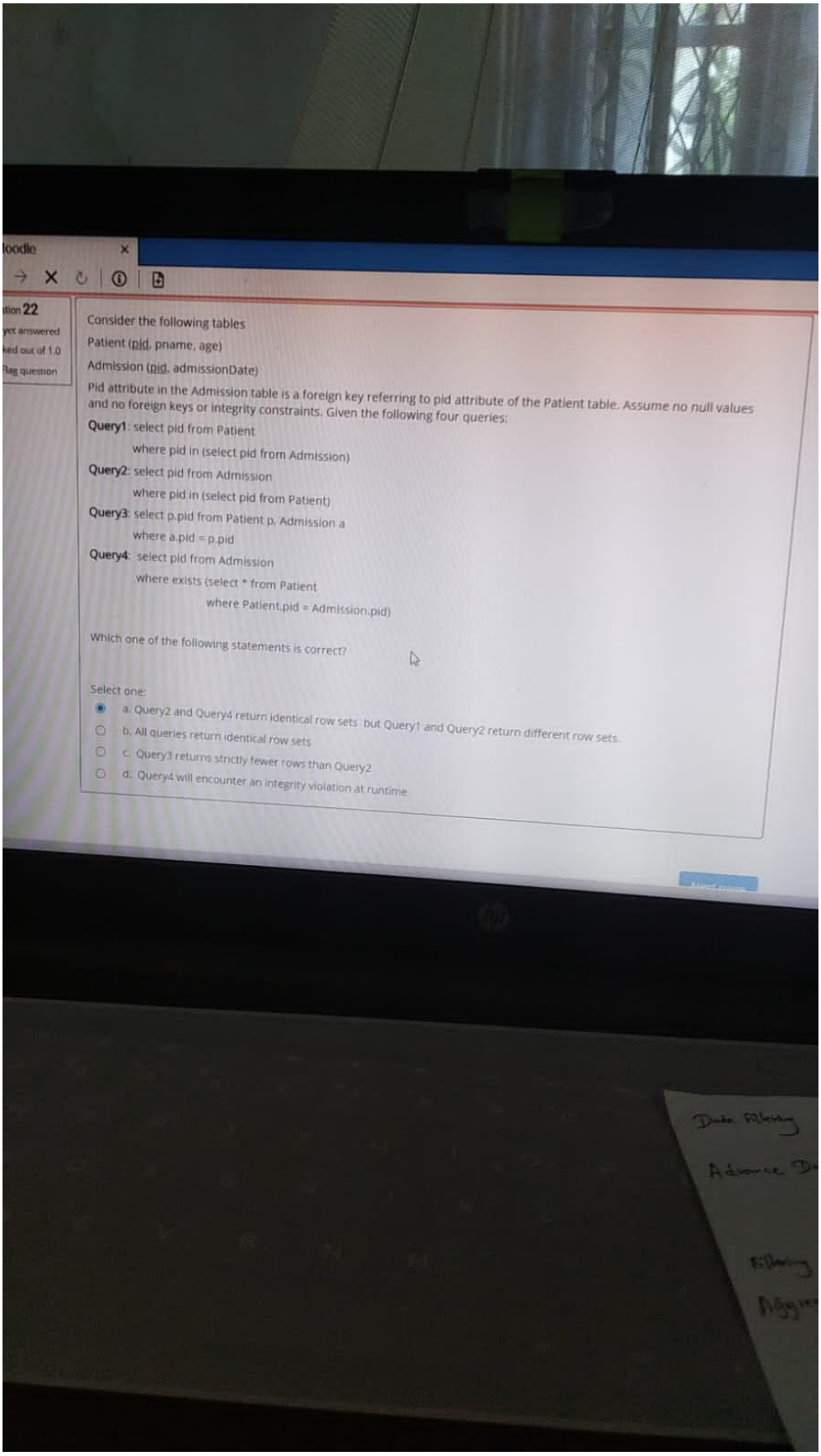
What will be the output of following SQL query ?

```
select * from emp e
where 2 = (select count(distinct e1.age)
           from emp e1
           where e1.age>e.age)
```

Select one:

- a. Third highest age
- b. Two distinct ages of employees
- c. Employee with second highest age
- d. Second highest age


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A required parameter (attempt) was missing
More information about this error
Continue


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ked out of 1.0
Flag question
Consider the following tables
Patient (pid, pname, age)
Admission (pid, admissionDate)
Pid attribute in the Admission table is a foreign key referring to pid attribute of the Patient table. Assume no null values and no foreign keys or integrity constraints. Given the following four queries:
Query1: select pid from Patient
where pid in (select pid from Admission)
Query2: select pid from Admission
where pid in (select pid from Patient)
Query3: select p.pid from Patient p, Admission a
where a.pid = p.pid
Query4: select pid from Admission
where exists (select * from Patient
where Patient.pid = Admission.pid)
Which one of the following statements is correct?
Select one:
a. Query2 and Query4 return identical row sets but Query1 and Query2 return different row sets.
b. All queries return identical row sets
c. Query3 returns strictly fewer rows than Query2
d. Query4 will encounter an integrity violation at runtime

Consider the appointments table given below

Appointments

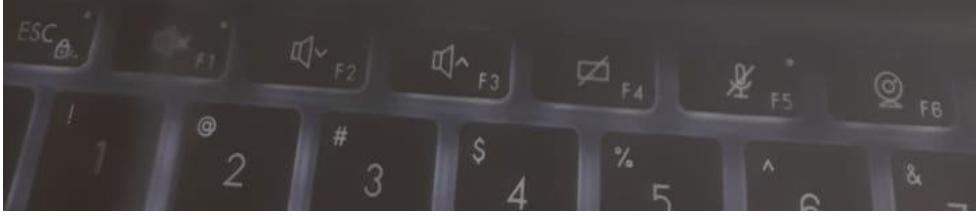
Patient	Doctor	appointmentDate
Lakmal	Dr. Janaka	08-01-2020
Nishani	Dr. Sunila	10-01-2020
Bhagya	Dr. Janaka	07-01-2020

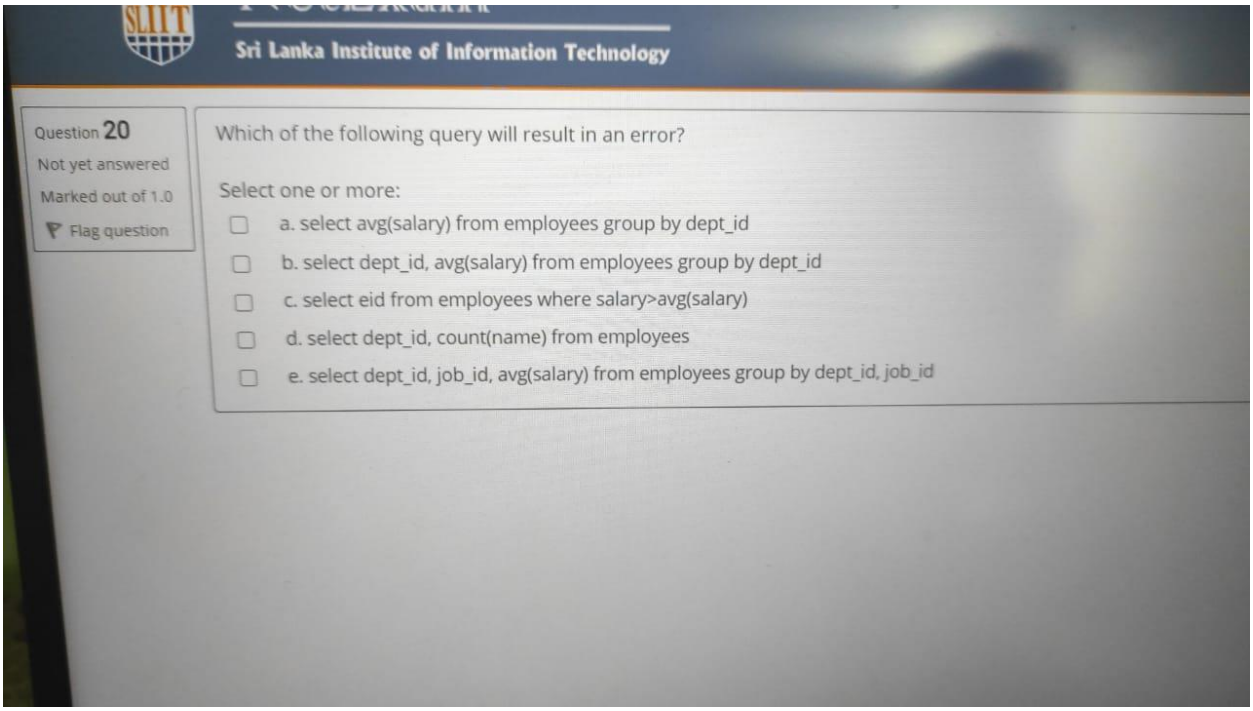
What is the output of the following SQL query?

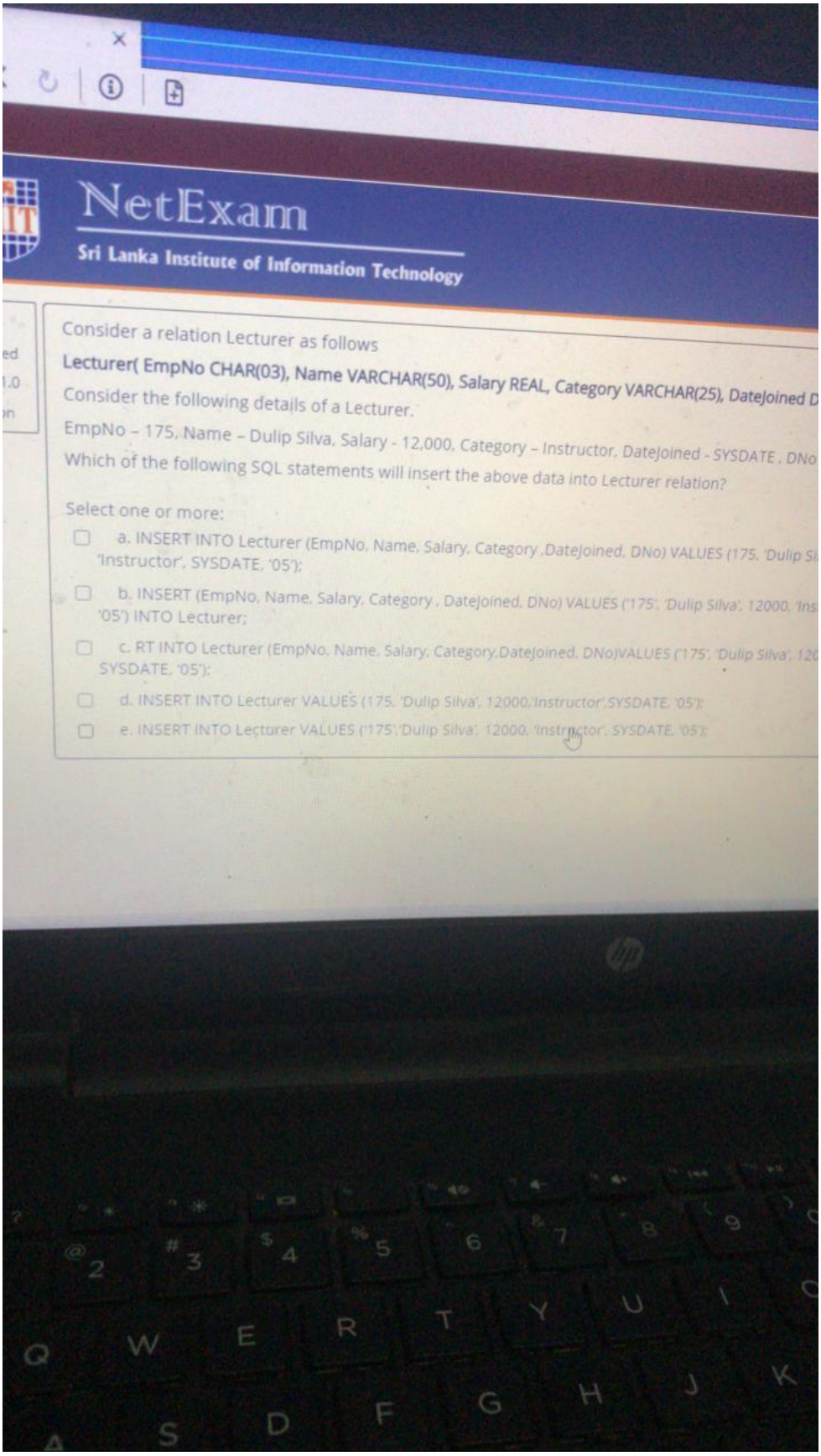
```
SELECT Count(*)
FROM  ( ( SELECT Patient, Doctor
          FROM Appointments) AS S
          INNER JOIN ( SELECT Doctoer, appointmentDate
                       FROM Appointments) AS T );
```

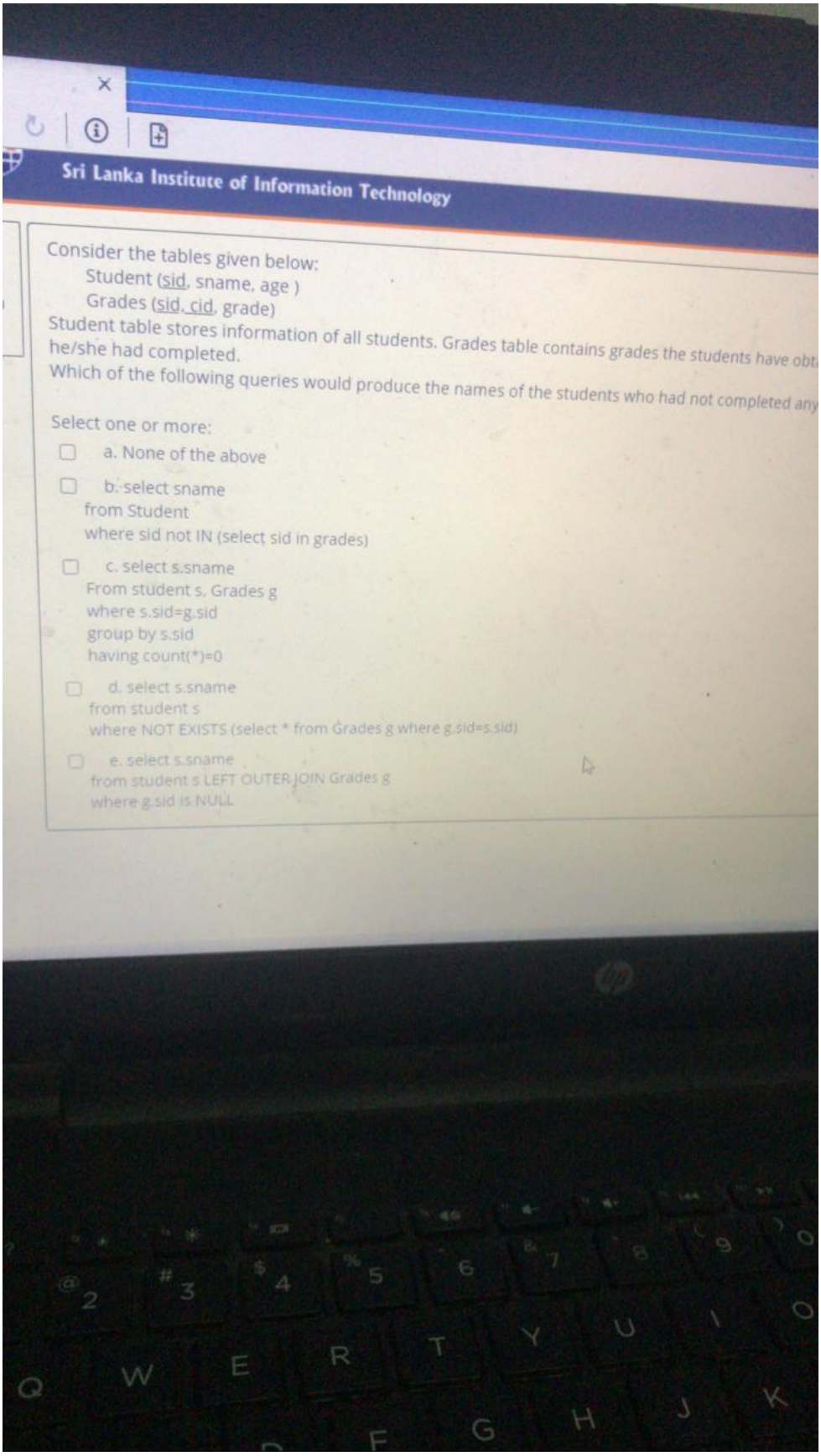
Select one:

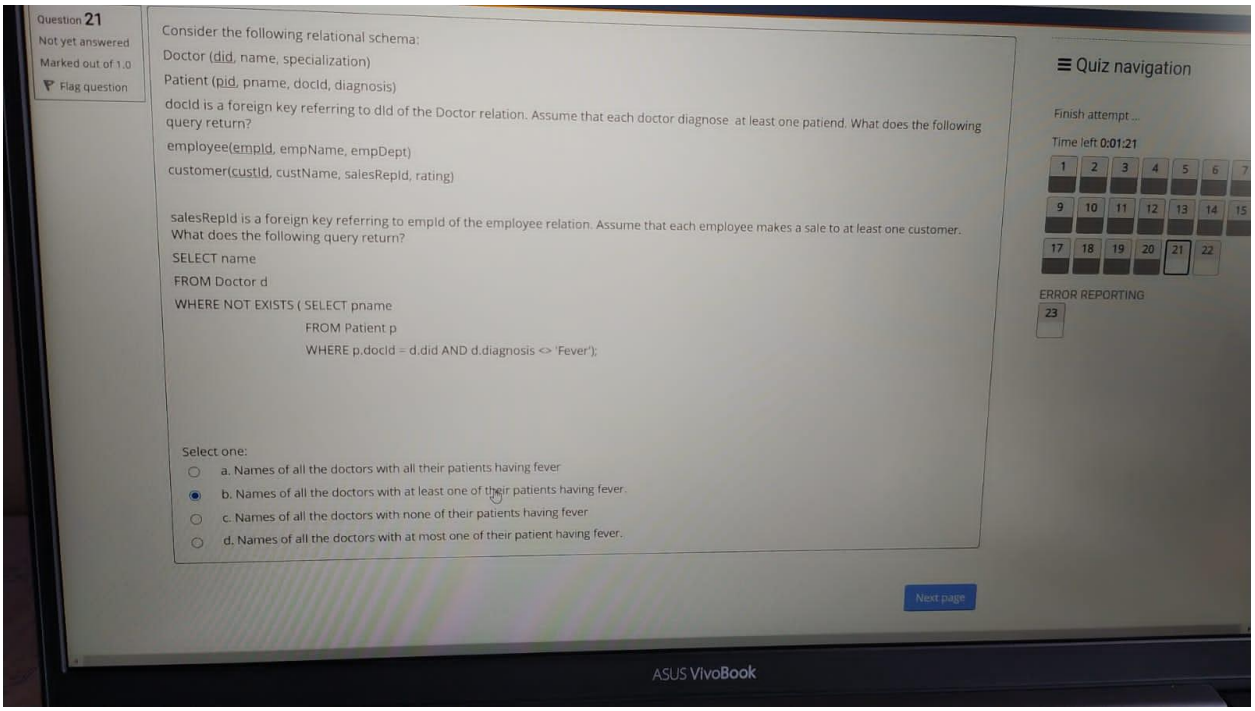
- a. 3
- b. 9
- c. 5
- d. 6




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Question 20
Not yet answered
Marked out of 1.0
Flag question
Which of the following query will result in an error?
Select one or more:
a. select avg(salary) from employees group by dept_id
b. select dept_id, avg(salary) from employees group by dept_id
c. select eid from employees where salary>avg(salary)
d. select dept_id, count(name) from employees
e. select dept_id, job_id, avg(salary) from employees group by dept_id, job_id


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Consider a relation Lecturer as follows
Lecturer(EmpNo CHAR(03), Name VARCHAR(50), Salary REAL, Category VARCHAR(25), DateJoined D
Consider the following details of a Lecturer.
EmpNo – 175, Name – Dulip Silva, Salary - 12,000, Category – Instructor, DateJoined - SYSDATE , DNo
Which of the following SQL statements will insert the above data into Lecturer relation?
Select one or more:
a. INSERT INTO Lecturer (EmpNo, Name, Salary, Category ,DateJoined, DNo) VALUES (175, 'Dulip Si
'Instructor', SYSDATE, '05');
b. INSERT (EmpNo, Name, Salary, Category , DateJoined, DNo) VALUES ('175', 'Dulip Silva', 12000, 'Ins
'05') INTO Lecturer;
c. RT INTO Lecturer (EmpNo, Name, Salary, Category,DateJoined, DNo)VALUES ('175', 'Dulip Silva', 120
SYSDATE, '05');
d. INSERT INTO Lecturer VALUES (175, 'Dulip Silva', 12000,'Instructor',SYSDATE, '05');
e. INSERT INTO Lecturer VALUES ('175','Dulip Silva', 12000, 'Instructor', SYSDATE, '05');


Sri Lanka Institute of Information Technology
Consider the tables given below:
Student (sid, sname, age)
Grades (sid, cid, grade)
Student table stores information of all students. Grades table contains grades the students have obt
he/she had completed.
Which of the following queries would produce the names of the students who had not completed any
Select one or more:
a. None of the above
b. select sname
from Student
where sid not IN (select sid in grades)
c. select s.sname
From student s, Grades g
where s.sid=g.sid
group by s.sid
having count(*)=0
d. select s.sname
from student s
where NOT EXISTS (select * from Grades g where g.sid=s.sid)
e. select s.sname
from student s LEFT OUTER JOIN Grades g
where g.sid is NULL


Question 21
Not yet answered
Marked out of 1.0
Flag question
Consider the following relational schema:
Doctor (did, name, specialization)
Patient (pid, pname, docId, diagnosis)
docId is a foreign key referring to dId of the Doctor relation. Assume that each doctor diagnose at least one patiend. What does the following query return?
employee(empId, empName, empDept)
customer(custId, custName, salesRepId, rating)
salesRepId is a foreign key referring to empId of the employee relation. Assume that each employee makes a sale to at least one customer. What does the following query return?
SELECT name
FROM Doctor d
WHERE NOT EXISTS (SELECT pname
FROM Patient p
WHERE p.docId = d.did AND d.diagnosis <> 'Fever');
Select one:
a. Names of all the doctors with all their patients having fever
b. Names of all the doctors with at least one of their patients having fever.
c. Names of all the doctors with none of their patients having fever
d. Names of all the doctors with at most one of their patient having fever.
Next page
Quiz navigation
Finish attempt ...
Time left 0:01:21
1 2 3 4 5 6 7
9 10 11 12 13 14 15
17 18 19 20 21 22
ERROR REPORTING
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